CHAPTER XLI

THE NEWER 'MATTER'

The twofold nature of light as a light-wave and as a light-quantum is thus extended to electrons and, further, to atoms: their wave-nature is asserting itself more and more, theoretically and experimentally, as concurrent with their corpuscular nature. (481)
A. SOMMERFELD

The concepts of wave amplitude, electric and magnetic field strengths, energy density, etc., were originally derived from primitive experiences of daily life, such as the observation of water waves or the vibrations of elastic bodies. (215)
W. HEISENBERG

The problem of quantum theory centers on the fact that the particle picture and the wave picture are merely two different aspects of one and the same physical reality. (215)
W. HEISENBERG

To me it seems extraordinarily difficult to tackle problems of the above kind, as long as we feel obliged on epistemological grounds to repress intuition in atomic dynamics, and to operate only with such abstract ideas as transition probabilities, energy levels, etc. (466)
E. SCHRÖDINGER

. . . for visualization, however, we must content ourselves with two incomplete analogies—the wave picture and the corpuscular picture. (215)
W. HEISENBERG

Not every physicist is an epistemologist, and not everyone must or can be one. Special investigation claims a whole man, so does the theory of knowledge. (326)
E. MACH

The following chapter was written in 1928 and since then the newer quantum mechanics has been developed much further, proved enormously fruitful, and has been repeatedly supported by experiments. The literature on this subject is steadily accumulating, the most important classical memoirs by the originators of this new scientific trend have been collected into book form and are now easily accessible. There is also a large number of excellent technical, as well as non-technical presentations. On reading in December 1932 what I had written in 1928, I find that although from some aspects the presentation may be considered unsatisfactory and antiquated, yet the epistemological side of the older presentation remains valid. So it seems advisable to retain this chapter and add only a few further $\bar{A}$ suggestions.

It is known that practically all creative and constructive physicists, who have produced revolutionary and lasting works, were interested in epistemology. There are many physicists who know as much physics as an Einstein, for instance, yet Einstein remains quite unique and his work is to a large extent responsible for the present revolutionary developments of physics. The reason is simple. Einstein has corrected a long established epistemological fallacy, which can be expressed in my language as the rejection of the structural fallacy of elementalism in a limited yet very important field of physics. He also established and applied new fundamental epistemological principles, which is another

way of saying that he established new standards of evaluation in physics, as for instance, that we should never postulate entities which cannot possibly be observed, that the 'laws of nature' should be formulated in terms of generally invariant relations expressed in tensor equations , .

The weakness of the system of Einstein, resulting in many futile criticisms, lies in the fact that he eliminated elementalism in one vital region of physics, but he did not formulate the *general epistemological principle* of *non-elementalism*, which should be applied everywhere, daily life included. This he could not have accomplished without a still deeper enquiry into the mechanism of time-binding, which produces all science, and which leads to the discovery of the fundamental fallacy in the use of the 'is' of identity. Only after the elimination of this remainder in us of the primitive man, does *structure* become the only possible link between the objective and verbal worlds, and becomes also the only possible content of 'knowledge'. 'Similarity of structure' then demands the complete and general elimination from science and life of any elementalism.

The strength of the newer quantum mechanics lies in the fact that the younger physicists have accepted the new epoch-making einsteinian standards of evaluation or epistemological principles; the weakness lies in the fact that the scientists do not realize that the fallacy of elementalism is entirely general and vitiates *all* scientific outlooks. No one can produce satisfactory theories, nor evaluate, nor interpret them properly as long as he continues to use the few-valued and *elementalistic* 'logics' and 'psychologies', which are at present always found at the bottom of any 'evaluation' or 'interpretation'.

The latest work of Dirac goes very far in the direction of building $\bar{A}$ physics by establishing his language of transformations, states, observables . , ascribing *structure* to protons, magnetic poles . , but even Dirac does not seem to realize the general fundamental $\bar{A}$ issues involved. Dirac says: 'The description which quantum mechanics allows us to give is *merely* a manner of speaking which is of value in helping us to deduce and to remember the results of experiments and which never leads to wrong conclusions. *One should not try to give too much meaning to it.*'[1] (Italics are mine.) The italicized words show that even Dirac does not realize fully the mechanism of identification, as otherwise he would not have used these words in this form. If we entirely abandon identification, then a theory or a book, being verbal, represents nothing else but special language; there is no 'merely' about that either, structure being the only possible link between the non-verbal and verbal worlds. Instead of warning the reader 'that one should not try to give too much meaning to it', we must simply *insist* that the *only* 'meaning' should be looked for in structure . , the 'too much meaning' always indicating inappropriate evaluation and ultimately semantic disturbances.

Current physical literature shows that the main problems of 'interpretation' depend on the *solution* of the *m.o* problems of 'observation', 'reality', 'fact'. , and border on the *scientific* solution of the problems of pathological 'delusions', 'illusions', and 'hallucinations', all of which involve the fundamental issues of the elimination of *identification*. But once the $\bar{A}$ issues are structurally formu-

lated and applied in practice, they result in a $\bar{A}$, *non-el*, ∞-valued orientation which involves the recognition of the *multiordinality* of terms . , which also solves the problems of quantum 'interpretations', the details of which I cannot enter into here.

Originally the quantum writers had an inclination to ascribe 'physical meaning' to waves. The present tendency of specialists is to regard the waves as 'purely symbolic', forgetting that experimentally something else besides the symbols 'bends around the corners'. From a $\bar{A}$ point of view, when the problems of the multiordinality of such terms as 'observation', 'fact', 'reality'. , are understood we will have to ascribe 'physical reality' to the waves, ascribe finer *structure* to the 'electron', . We would also have to abandon the A 'particle'-orientation and treat the 'electron', 'proton'. , in a $\bar{A}$, ∞-valued way as minute *fields*, which under the present experimental conditions behave as 'particles'. This $\bar{A}$ field-orientation suggests a great many possible interpretations, impossible in the A 'particle'-orientation.

Mathematically, the geometry of 'space filling' curves would have to be elaborated further so that we would better understand the *structure of plenums*, and this knowledge should be applied to physics.

We should also perform a direct series of experiments with a more elaborate Faraday box. A small wooden laboratory should be isolated from the rest of the world by every available *energetic screen* and physical experiments repeated under such new dynamic conditions. Technically the winding of, say, a foot thickness of insulated wires for different currents would not present any difficulties except for the door which should be also in the circuit. The eventual probable results of experiments in such a laboratory under different conditions could be calculated in advance, and it may be fairly well anticipated that at least significant discrepancies between the calculations and actual experiments would appear, throwing new light on the structure of the space-time plenum, the eventual connection between gravitation and electromagnetism , . Arguments alone will not help in this field and only experiments will point the pathway.

Because of the extensive literature dealing with the new quantum mechanics it seems unnecessary to dwell upon it any further, except by expressing the hope that some mathematicians and physicists will master the $\bar{A}$ ∞-valued orientation and will revise the theories now existing.

Section A. Introductory.

The new researches in the structure of the materials of the universe proceeded under unique conditions. On the one hand, since Planck in 1900 originated the quantum theory, which earlier elaboration we now call the classical quantum theory, the amount of experimental facts pointing in the direction of *some* quantum theory had become very convincing, yet, on the other hand, the lack of a structurally satisfactory theory to co-ordinate these new experimental facts was becoming distressing.

It appeared as though there were either a lack of 'geniuses' able to produce the required new theories, or the geniuses existed, but were unable to function properly. They seemed to suffer from a semantic blockage due to some identification which successfully prevented broader and unhampered vision.

Presently the co-ordinating theories *were* produced. In the production of the new quantum mechanics we see at work the unconscious semantic liberating influence of the $\bar{E}$ and $\bar{N}$ systems which had been developed and which we have already analysed. The classical theories had come to a structural impasse, but with the advent of younger scientists, who were educated in the theoretical semantic freedom of these new systems, the semantic blockage due to ascribing 'objectivity' to 'matter', 'space', and 'time' was removed, creative forces were released, and these young scientists proceeded to construct the structural formulations needed. They are now rightly hailed as geniuses.

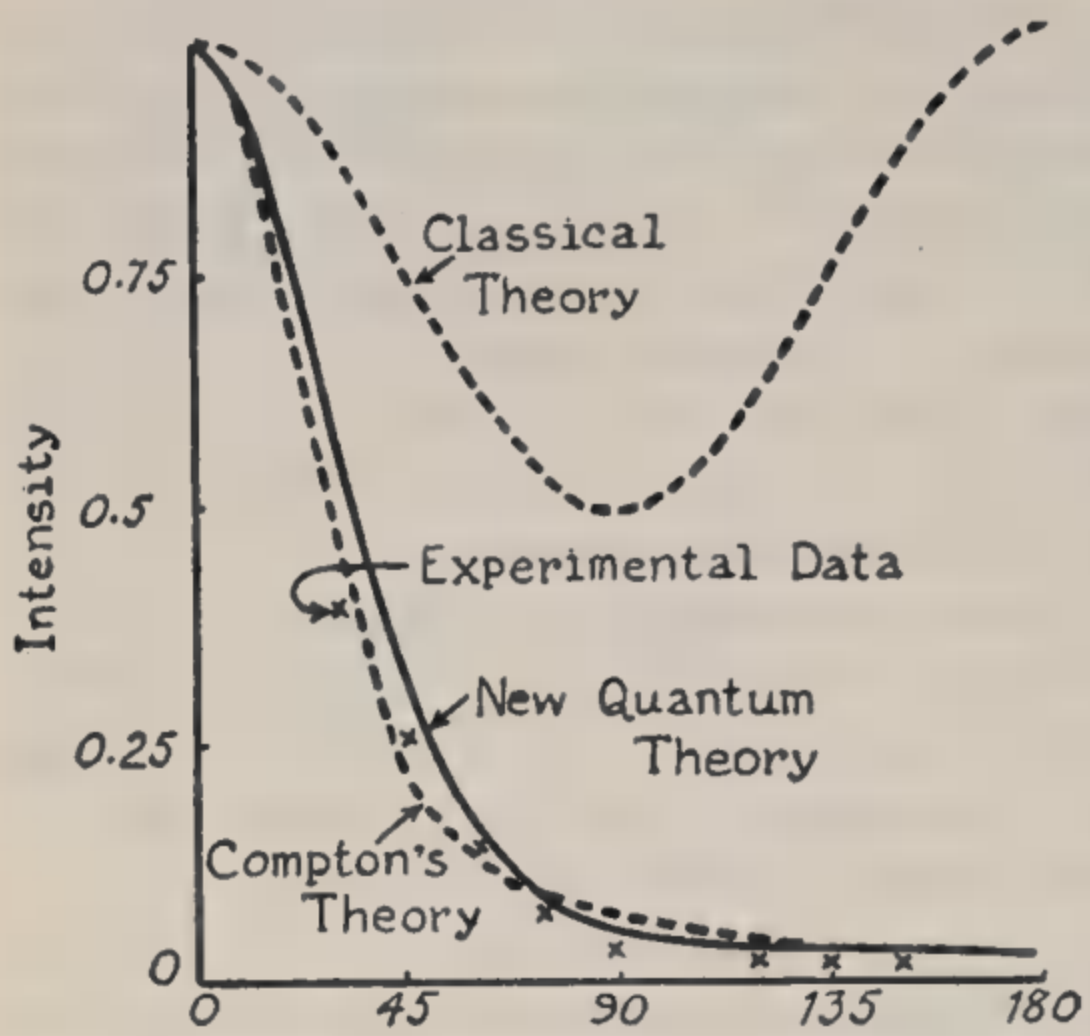
It is astonishing how these young post-einsteinian scientists of various lands and different inclinations produced, independently and practically simultaneously, various new quantum theories, using different methods and different mathematical, as well as national, languages. When these different theories were studied and compared they were found to amount practically to one theory, but expressed by different mathematical languages. Now the use of different mathematical languages to express one group of experimental facts brings an additional benefit in that it gives diversified verbal structural information about the problems at hand. Since these developments are very recent and progress has been extremely rapid it is hard to keep track of the status of the problem.

In this account of the newer quantum mechanics I will emphasize only the structural and semantic side, treating the different theories as behaviour of their respective authors, and as illustrating the issues above mentioned. From this point of view, we are not interested in discussing to what extent the given theories are 'true' or 'false', which means no more than similar or dissimilar to the world in structure. We are interested directly in those semantic aspects of human behaviour which have been neglected. When 'Smith' puts a black mark on white paper, it is to be called *human behaviour*, and behaviour *unique* for man. Our analysis does not involve the question of the validity of his doings. But since he did it, let us analyse his doings.

In speaking about theories as complex and technical as the newer quantum mechanics, it is practically impossible to give a satisfactory account in a non-technical way. Such theories have not yet been worked out thoroughly enough, many points are still not clear, and no proper evaluation is possible at present. These difficulties are really immaterial to our present purpose, because these theories are *empirical facts on record*, and they throw significant light on human behaviour. It is not here proposed to make reflections on the world around us, but to analyse a certain linguistic, structural form of human behaviour.

From the point of view of structure and *s.r*, the classical quantum mechanics would be quite enough for *sanity* and adjustment; it is enough to realize that the still older theories of 'matter', 'space', and 'time' are *el, structurally fallacious* and represent only primitive identifications.

The advent of such a crop of geniuses and of several theories expressed quite differently, yet nearly equivalent, is an event of deep human semantic significance. It helps to understand the working of the human nervous system, and is in accord with the present general theory.



(After *The New Quantum Mechanics* by G. Birtwistle, p. 192)

FIG. 1

From the point of view of the physicist, these new theories are a marked structural improvement over the classical theory, a fact which can best be illustrated by a diagram of a special case. In Fig. 1, the crosses indicate the experimental data, the curves indicate the results as predicted by the classical theory, by the Compton theory, and by the new quantum mechanics. It should be noticed that the new quantum theory appears much more in accordance with the experimental data than the older theories. This fact is of great structural and semantic importance to us as well as to the physicist.

Section B. The nature of the problem.

At this point we may explain briefly the nature of the problem that was demanding solution.

We have become familiar with the use of co-ordinates. This procedure has been generalized and has given rise to 'generalized co-ordinates'. These are defined as *arbitrary* variables which represent not merely lengths but may also represent angles, surfaces, volumes . , though they must be capable of representing the $3n$ orthogonal co-ordinates. If, in a special case, we make the number of generalized co-ordinates equal to the number of degrees of freedom which the system has, these s generalized co-ordinates can be regarded as independent of each other. If we denote by q_i the generalized co-ordinate, $q_i\ (i = 1 \text{ to } s)$, then the orthogonal co-ordinates of any of the n particles can be represented as definite functions of the generalized co-ordinates, so that

$$x_h = f_h\ (q_1, q_2, q_3, \ldots q_s)\ . \qquad (1)$$

We know that kinetic energy is represented by $mv^2/2$ where m represents the mass and v the velocity, or the 'time' derivative of the 'space' travelled.

If we want to find the value for the energy we must differentiate each of the $3n$ equations (1) with respect to 'time', which gives the components of the velocity, square them, multiply them by the corresponding masses and add them together to find the double value for the energy.

For simplicity we will denote the 'time' derivatives by the chosen letter, but with a dot over it (newtonian method). Thus,

$$\dot{x}_h = \frac{\partial x_h}{\partial q_1}\dot{q}_1 + \frac{\partial x_h}{\partial q_2}\dot{q}_2 + \cdots + \frac{\partial x_h}{\partial q_s}\dot{q}_s \tag{2}$$

Squaring (2), we have

$$\left.\begin{aligned} x_h^2 = {} & \left(\frac{\partial x_h}{\partial q_1}\right)^2 \dot{q}_1^2 + \left(\frac{\partial x_h}{\partial q_2}\right)^2 \dot{q}_2^2 + \cdots + \left(\frac{\partial x_h}{\partial q_s}\right)^2 \dot{q}_s^2 \\ & + 2\frac{\partial x_h}{\partial q_1}\frac{\partial x_h}{\partial q_2}\dot{q}_1\dot{q}_2 + 2\frac{\partial x_h}{\partial q_1}\frac{\partial x_h}{\partial q_3}\dot{q}_1\dot{q}_3 + \cdots \\ & + 2\frac{\partial x_h}{\partial q_2}\frac{\partial x_h}{\partial q_3}\dot{q}_2\dot{q}_3 + \cdots \end{aligned}\right\} \tag{3}$$

The last expression (3) can be simplified:

$$\dot{x}_h^2 = \sum_{i=1}^{i=s}\sum_{k=1}^{k=s}\frac{\partial x_h}{\partial q_i}\frac{\partial x_h}{\partial q_k}\dot{q}_i\dot{q}_k . \tag{4}$$

It is easy to see that if in (4) we put $k=i$ we will have square members, and when $i \neq k$, every term will occur twice and so the above abbreviation (4), covers the formula (3).

If we write similar expressions for all of the $3n$ orthogonal co-ordinates, multiply by the corresponding masses and then add them together we obtain twice the value for the kinetic energy $2L$.

$$2L = \sum_{i=1}^{i=s}\sum_{k=1}^{k=s} c_{ik}\,\dot{q}_i\,\dot{q}_k, \text{ where}$$

$$c_{ik} = \sum_{h=1}^{h=n} m_h\left[\frac{\partial x_h}{\partial q_i}\frac{\partial x_h}{\partial q_k} + \frac{\partial y_h}{\partial q_i}\frac{\partial y_h}{\partial q_k} + \frac{\partial z_h}{\partial q_i}\frac{\partial z_h}{\partial q_k}\right].$$

The coefficients in the expansion of c_{ik} depend only on the values of the generalized co-ordinates and are independent of the value of the time-derivatives. The time-derivatives can be properly called *generalized velocities*, and we may denote them by $q_i.^2$

In establishing formulae for the quantum theory we want to be as general as possible and not restrict ourselves to vibrational energy only. But we want to take into consideration *any* arbitrary point-mass, independently of whether we assume this point to be charged or not.

We define the momentum or impulse as the product of the mass and the velocity, or, $p=mv$. If, instead of denoting our co-ordinates by x, y, and z, we use the generalized co-ordinates q_i, we would have for the magnitude and direction of the velocities the time-derivatives of the co-ordinates; namely, $\dot{q}_i$, where $\dot{q}_1=\dot{x}=dx/dt$, $\dot{q}_2=\dot{y}=dy/dt$, .

If p_1, p_2, p_3, represent the corresponding components of the momentum or impulse, then we would have $p_i=m\dot{q}_i$. (5)

We should notice that the dynamical triplet of impulse co-ordinates occurs conjointly with the geometrical triplet of the co-ordinates of position. The second law of motion tells us that 'the change in momentum is proportional to the impressed force and takes place in the direction in which that force acts'. If we assume that the force K is derivable from the potential energy E_{pot}, (a function of q_i), then we have $\dot{p}_i = K_i = \dfrac{-\partial E_{pot}}{\partial q_i}$. (6)

The kinetic energy (E_{kin}) is represented by

$$E_{kin} = \frac{m}{2}(\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2) = \frac{p_1^2 + p_2^2 + p_3^2}{2m}$$

where by (5), $\dot{q}_i^2 = p_i^2/m^2$. We call the total energy, which is represented as the sum of the kinetic and the potential energy, as expressed in terms of the generalized co-ordinates and momenta, the hamiltonian function H. Then we have:

$$H(q, p) = E_{kin} + E_{pot}, \quad \frac{\partial H}{\partial q_i} = \frac{\partial E_{pot}}{\partial q_i}, \quad \frac{\partial H}{\partial p_i} = \frac{\partial E_{kin}}{\partial p_i} = \frac{p_i}{m}. \tag{7}$$

From (5), (6), and (7), we get the fundamental equations of motion,

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \text{ and } \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}.$$

The above hamiltonian, or canonical, form of the equations is remarkable because it preserves its form if any arbitrary co-ordinates are introduced; it is invariant under the transformation of co-ordinates. The equations hold not only for an individual point-mass but also for any arbitrary mechanical system. For arbitrary co-ordinates and systems the momentum or impulse p is defined by

$$p_i = \frac{\partial E_{kin}}{\partial \dot{q}_i}, \quad \text{so that the}$$

kinetic energy is expressed as a function of the q_i's and their derivatives the $\dot{q}_i$'s.

To help visualization we can construct and consider the p and q as rectangular co-ordinates in two dimensions in the phase plane of our system. In this plane the sequence of those graph-points that correspond to the successive states of motion of the system represent the phase paths or phase-orbits. The characteristic structural feature of the quantum theory is that it selects a discrete family of phase-orbits from the infinity of possible orbits.

We next consider a point-mass m that is bound elastically to its position of rest, and which can move to either side of the central position only in the direction $x = q$, or its reverse, when experiencing a restoring force. We call such point-mass a linear oscillator. If we wish to emphasize that our oscillator is capable only of definite vibrations, on account of its elastic attachment, we call it a 'harmonic oscillator'. If the vibration number, or the frequency of the oscillator, which is represented by the number of its free vibrations per unit of 'time', is denoted by ν, then the vibration is represented by $x = q = a \sin 2\pi\nu t$.

The impulse becomes $p = mv = mq = 2\pi\nu ma \cos 2\pi\nu t$. The phase-orbit is represented by an ellipse in the p-q-plane and is given by the equation $\frac{q^2}{a^2} + \frac{p^2}{b^2} = 1$, where the minor axis $b = 2\pi\nu ma$.

In our family of orbits the phase area between two orbits is equal to the quantum of action h. Sommerfeld regards h as an elementary region or element of the phase area, and considers it as the definition of the Planck quantum of action h. If W_n represents the energy of the oscillator when it describes the n-th orbit, then $W_n = nh\nu$. In these orbits the energy appears as a whole multiple of the elementary quantum of energy; $\epsilon = h\nu$, and $W_n = n\epsilon$.

We call *stationary states* of the oscillator those states which the oscillator may pass through without cessation and without loss of energy, or, without radiation.

When an oscillator retains its stationary state, its energy is constant and its graph appears as an ellipse of the family in the phase plane. However, when the energy of the oscillator changes and jumps over to a smaller orbit, it emits energy. When it passes to a larger orbit it absorbs energy. The emission and absorption of energy occurs in multiples of the energy quantum, ϵ.

The graphs of the system in the phase plane are restricted to certain 'quantised' orbits. Between each orbit and its successor there is an elementary region, of area h. The n-th orbit, if closed, has as area nh. Or, expressed symbolically, $\int p dq = nh$. This integral is called the phase integral and is taken along the n-th orbit.

The quantum hypothesis can be structurally formulated so that the phase integral must be a whole multiple of the quantum of action h. This form of the classical quantum postulate is more general than the original formulation of Planck, although it includes the latter as a particular case.

In case of a rotating point-mass, a similar analysis gives us $E_{kin} = \frac{p\dot{q}}{2}$ and when $\nu = \frac{q}{2\pi}$, $E_{kin} = \frac{nh}{2}\frac{\dot{q}}{2\pi} = \frac{nh\nu}{2}$ where ν represents the rotation frequency of the rotator, or the number of full revolutions per unit of 'time', and takes the place of the vibration number of the oscillator.

In the classical theory, the quantised states were distinguished from all other possibilities by the characteristic whole numbers, and so we had a network. In a quantum orbit the 'electron', if undisturbed, was supposed to move permanently without resistance and not to emit radiation. The phase-space, representing the manifold of the possible states, including non-stationary states, is crossed, mesh-like, by the graph curves of the stationary orbits. The size of the meshes is determined by Planck's constant h.[3]

Section C. Matrices.

The older quantum mechanics forms an elaborate system, and we have a large accumulation of numerical data on record. Some of these data corroborated the older theories nicely, but some data were in contradiction to the classical

theory. The problem was not to discard the numerical data, which, whatever they mean structurally, represent quite solidly established data, but to find new equations which would be satisfied by these facts. Now 'new equations' really mean *languages of new structure*, and therefore new formulations had to be discovered.

Dealing with tables which give special theoretical data, it was natural to start with a calculus which deals with such numerical special tables. Such a calculus had been developed long ago, and was called the matrix calculus. Later on, when matrices themselves were treated as complex quantities, and still later, as operators, we were enabled to pass to the more developed calculi which use ordinary differential equations. The new quantum theories give us a unique case, in which several mathematical methods have been used at once and of which the results are fairly in accord.

At this point it is advisable to give a few structural explanations of these mathematical notions, including the matrix calculus. If we have two equations of the first degree with two variables; namely, $a_1x+b_1y=c_1$ and $a_2x+b_2y=c_2$, the solution of these equations takes the form: $x=\dfrac{b_2c_1-b_1c_2}{a_1b_2-a_2b_1}$; $y=\dfrac{c_2a_1-c_1a_2}{a_1b_2-a_2b_1}$

The common denominator of the two solutions can be written in a two-dimensional table

$$\begin{vmatrix} a_1, & b_1 \\ a_2, & b_2 \end{vmatrix} \tag{1}$$

which is understood as the product of the upper left-hand number and the lower right-hand number, minus the product of the lower left-hand and upper right-hand numbers.

Similarly, the numerators of these solutions can also be represented in the form of two-dimensional tables; namely,

$$b_2c_1-b_1c_2=\begin{vmatrix} c_1, & b_1 \\ c_2, & b_2 \end{vmatrix} \quad (2) \text{ and } \qquad c_2a_1-c_1a_2=\begin{vmatrix} a_1, & c_1 \\ a_2, & c_2 \end{vmatrix} \tag{3}$$

to which the above-mentioned rule applies. Expressions like (1), (2), (3), are called determinants of the second order.

The numbers in the first, second . , horizontal lines are called the first, second . , *rows*, respectively; the vertical lines are called first, second . , *columns*.

The above definitions and method can be applied to any number of equations with an equal number of variables, and in each case our determinant would have n^2 numbers, n rows and n columns.

We may use another notation which employs one letter for the coefficients of our variables, with indexes or suffixes to indicate that their values are different. Let us consider n^2 elements in the table:

$$\begin{vmatrix} a_{1,1}, & a_{1,2}, & a_{1,3}, & \ldots, & a_{1,n} \\ a_{2,1}, & a_{2,2}, & a_{2,3}, & \ldots, & a_{2,n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n,1}, & a_{n,2}, & a_{n,3}, & \ldots, & a_{n,n} \end{vmatrix}. \tag{4}$$

The expression (4) is called a determinant of the n-th order.

The notation by suffixes is very convenient and is very much used these days. The first suffix denotes the row, the second the column in which the element is situated. Usually the comma dividing the two numbers in the index is dropped and the coefficients are written simply a_{11}, instead of $a_{1,1}$. In general the element $a_{i,k}$, or a_{ik}, represents the element in row i and column k.

The elements lying in the diagonal joining the upper left-hand to the lower right-hand number are called the principal diagonal. In our example we notice that the elements in the diagonal are such that $i=k$.

We have definite rules by which we can arrive at the solution of our equations, once the coefficients, which are the elements of the determinant, are given. In general, the determinants are treated as a functional form.

If m and n are positive integers, a manifold, or system of mn ordered quantities or elements arranged in m horizontal and n vertical rows, will be called a *rectangular matrix* and we may use the notation $A=(a(mn))$:

$$\begin{pmatrix} a(11) & a(12) & a(13) & \ldots\ldots \\ a(21) & a(22) & a(23) & \ldots\ldots \\ a(31) & a(32) & a(33) & \ldots\ldots \\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots \end{pmatrix}, \text{ or } (a(nm)).$$

The numbers m and n are called the orders of the matrix. If $m=n$ the matrix is called a *square* matrix. Without loss of generality we can treat any rectangular matrix in which $m \neq n$ as a square matrix by supplementing the missing rows and columns with zeros.

A matrix of the type,

$$1=\begin{pmatrix} 1 & 0 & 0 & 0 & .... \\ 0 & 1 & 0 & 0 & .... \\ 0 & 0 & 1 & 0 & .... \\ 0 & 0 & 0 & 1 & .... \\ .. & .. & .. & .. & .... \end{pmatrix} \text{ or } (\delta(nm)),$$

where $\delta(nm)=1$ for $n=m$, and $\delta(nm)=0$ for $n \neq m$, is called a unit matrix.

The matrix

$$\begin{pmatrix} a(11) & 0 & 0 & \ldots\ldots \\ 0 & a(22) & 0 & \ldots\ldots \\ 0 & 0 & a(33) & \ldots\ldots \\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots \end{pmatrix} \text{ or } (a(nm)\ \delta(nm)),$$

is called a diagonal matrix. In the new quantum mechanics a diagonal matrix is independent of t and represents a constant of the classical theory. The reverse is not necessarily true. The operation of differentiation can be expressed in terms of multiplication of matrices with the aid of the unit matrix.[4]

Equations in which matrices are equated are called matrix equations. If the equations involve only one unknown matrix, which does not occur more than once as a factor, such equations are called matrix equations of the first degree.

The m scalar equations

$$\begin{aligned} a_{11}x_1+a_{12}x_2+\ldots+a_{1n}x_n&=c_1\\ a_{21}x_1+a_{22}x_2+\ldots+a_{2n}x_n&=c_2\\ \cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot&\\ \cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot\quad\cdot&\\ a_{m1}x_1+a_{m2}x_2+\ldots+a_{mn}x_n&=c_m \end{aligned}$$

are together equivalent to one single matrix equation. There are several ways in which the notation can be simplified.

The difference between a determinant and a matrix is subtle, but important. By a determinant we understand, by definition, a certain homogeneous polynomial of the n-th degree, in the n^2 elements a_{ij}. Accordingly, a determinant gives a definite number when calculated.

But in many instances we are interested in the *table*, or the n^2 elements arranged in a certain order but *not* combined into a polynomial. Such an array, or table, is called a matrix. Thus, from this point of view, a matrix does not represent a definite quantity, but a system of quantities, and so a matrix is *not* a determinant.

We can illustrate this difference by an example. If we take a determinant of the second order

$$\begin{vmatrix} a_1, & b_1 \\ a_2, & b_2 \end{vmatrix}$$

and change the rows into columns, or vice versa, thus:

$$\begin{vmatrix} a_1, & a_2 \\ b_1, & b_2 \end{vmatrix}$$

the value of both determinants will be equal; namely,

$$a_1b_2-a_2b_1,$$

by the definition rule already given; yet the *matrices* of the two determinants are *different.*

Although different, a determinant nevertheless defines a matrix, called the matrix of the determinant; conversely, a matrix defines a determinant, called the determinant of the matrix.

We have said that a matrix does not represent a quantity, while a determinant does. At this stage, and from this point of view, we may say so legitimately. However, we might eventually treat a matrix as a quantity also; but for this purpose we should have to enlarge the meaning of the term 'quantity'.

In our present use of the term 'quantity' we mean the real and complex quantities of ordinary algebra.

It may be said that mathematicians have had a peculiar tendency, which has proven of great value in the development of mathematics, gradually to extend the meaning of terms in order to embrace new notions as they arise. For instance, we have enlarged the primitive meaning applied to positive integers to embrace negative numbers, which formerly would not have been considered as quantities. Similarly, if we here use the ordinary notion of algebraic quantity, then a matrix is not a quantity but a system of quantities. The problem is, how shall we enlarge this meaning to include the matrices?

Mathematics recognizes that this generalization of *mathematical* notions is extremely useful and *legitimate*. This structural issue appears to be of very general application, as all of us exhibit a tendency towards it. It is a purely mathematical and useful tendency in mathematics, but it leads to disastrous results when applied to daily-life abstractions, as explained in Part VII. In this connection we should recall the difference between the mathematical contentless abstractions and the abstractions with physical content, with which we are generally concerned in science and life.

Let us now follow up the method by which a matrix can be considered as a quantity. If we have objects of two or more kinds which can be counted or measured, and if we consider an aggregate of such objects, say 5 horses, 3 cows and 2 sheep, we could denote such a complex quantity by the symbol (5,3,2). In this case, the first place in our symbol would be reserved for horses, the second for cows, and the third for sheep.

In mathematics, we do not specify horses, or cows, or sheep, but consider sets of quantities, and distinguish them by the position which they have in our symbolism. We may denote such a complex quantity by a single letter, $A=(a,b,c,)$. (For instance, we denote a fraction by a single letter, although a fraction is specified by *two* numbers.)

In such an instance, we should call a complex quantity equal to another when, and only when, the components are respectively equal. And a complex quantity is said to vanish only in case all the components vanish.

Ordinary mathematical operations can be applied to such complex quantities. For instance, we may define a sum or difference of two complex quantities

$$A'=(a_1, b_1, c_1) \text{ and } A''=(a_2, b_2, c_2) \text{ as}$$
$$A' \pm A'' = (a_1 \pm a_2, b_1 \pm b_2, c_1 \pm c_2) \ . \ ,$$

a definition which is entirely satisfactory theoretically, and also practically, as can be verified from our example.

From this point of view we may consider a matrix as a complex quantity with mn or m^2 components. A matrix would then represent a complex quantity, as a special case under the general method sketched above.

We could then define our further operations. A matrix would be said to be zero when all elements are equal to zero. Two matrices would be said to be equal when they have equal numbers of rows and columns and every element of one is equal to the corresponding element of the other.

By setting up some such rules we could develop a calculus of matrices, and matrices would be considered as complex numbers. In general, the algebraic rules would be found to be applicable to matrices, which would further justify us in treating matrices as complex numbers.

One of the notable exceptions in our operations would be found in the application of the classical operation of multiplication and its dependencies. In ordinary algebra and arithmetic, multiplication is what is called 'commutative' which means that $2 \times 3 = 3 \times 2 = 6$, or $a \times b = b \times a$.

In defining the multiplication of matrices we have no *a priori* grounds for determining why one definition or restriction should be preferable to another.

Only practice can show which definition is more workable or more fruitful in results. In the matrix calculus the definition of Cayley is generally accepted, as it has led to the most workable results. It was based on considerations of the composition of *linear* tranformations.

The definition is approximately as follows: The product **ab** of two square matrices of the n-th order gives a square matrix of the n-th order in which the element which lies in the i-th row and j-th column is obtained by multiplying each element of the i-th row of **a** by the corresponding element of the j-th column of **b** and adding the results.

If we denote by a_{ij} and b_{ij} the elements in the i-th row and j-th column of **a** and **b** respectively, then by definition the element (i,j) of our product **ab** would give,

$$a_{i1}b_{1j}+a_{i2}b_{2j}+\cdots+a_{in}b_{nj}, \tag{5}$$

and the (i,j) element in the matrix **ba** would be

$$a_{1j}b_{i1}+a_{2j}b_{i2}+\cdots+a_{nj}b_{in}. \tag{6}$$

In general, the quantities (5) and (6) are not equal and therefore we see that the multiplication of matrices is, in general, *not commutative*. The *order* in which we perform our multiplication is of importance and **ab** *is not* generally equal to **ba**, ($\mathbf{ab} \neq \mathbf{ba}$).[5]

It should be noticed that the vector calculus has made us familiar with new operations which differ from arithmetical operations. For instance, the sum of two vectors differs in general from the arithmetical sum and is defined by the law of the parallelogram (see Chapter XXXIII). This definition is more general, and the arithmetical definition expresses only the particular case in which the vectors have one direction. Similarly, the non-commutative law of multiplication corresponds more closely to vector multiplication than to arithmetical multiplication.

We will not go further into the details of the matrix calculus, which is a well-developed mathematical discipline with a large literature, but will emphasize some methodological points of importance.

One of the main applications of the theory of matrices is found in the subject of *linear transformations*.

In mathematics, instead of using the given variables, we very often introduce new variables which are functions of the old. Such transformations, or change of variables, are particularly simple and important when the functions in question are homogeneous and *linear*.

If $x_1, x_2, \ldots, x_n$ represent the original variables, and $x_1', \ldots, x_n'$ the new variables, we have, by definition, the formulae of transformations:

$$\begin{aligned} x_1' &= a_{11}x_1+\cdots+a_{1n}x_n \\ &\cdots\cdots\cdots\cdots \\ x_n' &= a_{n1}x_1+\cdots+a_{nn}x_n\,. \end{aligned}$$

The square matrix made up of the coefficients is called the matrix of the transformation and the determinant is called the determinant of the transformation and is completely determined by the matrix. We have already seen the importance of linear equations and linear transformations in physics and therefore in

the investigation of the world around us. The theory of matrices is connected with such transformations, hence the importance of the theory of matrices for physics.

For our purpose another characteristic of the matrix calculus is of interest and that is the fact that in physics we usually have a large number of empirical numerical data which enter as coefficients in equations and which can always be put in the form of a two-dimensional array of numbers, or a table, which we have just called a matrix.

It appears that every physical quantity, however complicated, can be represented by such a table giving the values of the parameters which determine its character. From the definition of the term 'variable' as *any* value out of a possible range of values, we might treat our variables in two distinct ways, one from the point of view of *function* or operations, the other from the *extensional* point of view, when the function or operations are unknown, although the particular values of the variable are given. The matrix calculus takes this last point of view.

In physical research work we deal for the most part with arrays of numbers or unique and specific, mostly asymmetrical, relations which the experiments give us. Our usual problem is to find the structure, the function, and the operations which are satisfied by the given experimental relations.

We see that the dual approach to our solutions is due entirely to the definition which we have accepted for the variable. We have two issues: Either to find the values of the variable which satisfy the given function and operations or, having particular values of the variable (experimental), to find the function and operations.

Obviously every physical quantity can be represented by a matrix, which may be a sequence, and every mathematical theorem can be reduced to a property of matrices. Once the proper mathematical theories are worked out it will be always possible to pass from one form of representation to the other.[6]

In the older mechanics the functions were rather obvious and so the use of the matrix calculus was not so imperative. In the newer mechanics, the opposite is the case. We have a large amount of numerical experimental relations, but the functions and operations connecting these variables are unknown and the problem is to find them. From this point of view the matrix calculus represents an *extensional* calculus, a calculus of *observation.* In using the descriptive term 'observation' we must add that some objections have been advanced to such a use of the term. The answer is that the term 'observation', like most of our most important terms, must be considered a *multiordinal term.* Once this is understood the objections to the use of the term do not hold.

There is no limitation as to what the elements of a matrix may represent; they may be functions, functions of functions , .

Section D. The operator calculus.

The use of the *operator calculus* is interesting, structurally and psychologically, in that attention is concentrated, not on the numerical quantities,

but on the semantic *operations of combining* them. The calculi used in the newer quantum mechanics are peculiar, because, while they retain the numerical data, and as far as possible, the classical equations, they alter the operations by which these quantities are combined, or the interpretation of the equations.

As an illustration of such a procedure we can take two different formulae for the addition of velocities; one from the classical mechanics, where

$$V_{13} = V_{12} + V_{23}, \tag{1}$$

and the other, the formula for velocity as given by the Einstein theory; namely,

$$V_{13} = \frac{V_{12} + V_{23}}{1 + \frac{V_{12} \times V_{23}}{c^2}}. \tag{2}$$

In these formulae V_{12} represents the velocity of body 1 relative to body 2 , . In formula (1) the sign '+' symbolizes the ordinary arithmetical operation of addition. As we already know, this formula has proven too simple to represent accurately the experimental data, and Einstein has replaced it by the more elaborate formula (2).

The above statement is the usual way of speaking about the modification in formulation which has taken place in physics since Einstein. But we could equally well say that the formula has *not* been altered except that the '+' has no longer the old meaning and does not now represent the arithmetical operation of addition. Both points of view lead ultimately to one value, V_{13}, and the *computations* are similar in both cases.

We should notice especially the great freedom with which we can treat mathematical entities. Our voluntary selection of the point of view becomes important. A similar freedom of selection of interpretation appears to a still larger extent in all *verbal* problems, a fact of considerable structural and semantic importance in any theory of sanity, as we have already seen.

Further illustration of this freedom can be seen in the way in which the ordinary notion of multiplication is re-interpreted in the operator calculus. Let us denote by q and f, two numerical quantities, and qf as their product. But we could view this problem differently. We could say that qf *results* from a semantic *operation* q performed on f, or a semantic *operator* $(q \times)$ acting upon f which transformed f into qf. We could denote the *operation* of multiplying by q or the operator $(q \times)$ by a single symbol Q. Quite obviously the operator Q is not the number q; in other words, the semantic operation of multiplying, say by two, is *not* the number 2.

The operation of multiplying integer 1 by integer 2 gives the result 2. Similarly, the operation of multiplying 1 by q, or, in our new language, the application of the operator Q to the integer 1, gives q. In symbols, $Q1 = q$. If we take any arbitrary function f, the result of the operation of Q upon f is

written $Qf = qf$. If we were to follow the operation Q by the differential operation d/dx the result would be*:

$$\frac{d}{dx}(Qf) = \frac{dq}{dx}f + q\frac{df}{dx} = \left[\frac{dq}{dx}\times\right]f + Q\frac{df}{dx}.$$

But as f is arbitrary, it may be omitted from the equations and the result written in the operator form as

$$\frac{d}{dx}Q - Q\frac{d}{dx} = \left[\frac{dq}{dx}\times\right] = \frac{dQ}{dX}.$$

The symbol $\frac{dQ}{dX}$, defined by this equation, should be read as 'the operation of multiplying by dq/dx'. Similarly, $\frac{d(QP)}{dX} = \frac{dQ}{dX}P + Q\frac{dP}{dX}$.

In translating the ordinary equations into the language of operators nothing new is introduced. This translation involves only a change of *'mental' focus.* Instead of concentrating our attention on the numerical values we concentrate on the operations of combining them. Since the great problems of the quantum mechanics consist in finding new methods of computation, or of combining numerical values, such a change of attitude may prove to be structurally useful.

It should be noticed here that once matrices are considered, and treated, as quantities, or unique and specific relations, by similar reasoning they can be treated as operators. This problem is of fundamental structural and semantic importance because in the quantum theory we deal with matrices which have infinite numbers of terms and since this complexity presents great technical difficulties it is of enormous advantage to be able to pass to some more developed methods of calculation.

At the preliminary stage in the operator calculus we have assumed that multiplication was commutative, that $QP = PQ$; but in the further and more general development of the theory, this does not hold.

In general we must assume in the operator calculus that multiplication is not commutative, that $QP \neq PQ$. For instance, if $Q = (q\times)$ and $P = d/dq$, the two operations are certainly not commutative. Naturally, the validity of $2\times 2 = 4$ is not doubted, but generalized non-commutative multiplication has a definite asymmetrical and so structural geometrical interpretation, to be found in the vector calculus. When we associate with each numerical quantity its own operation of multiplication we thus obtain a more general calculus. Operators may be regarded as compound, or built up from the elementary arithmetical operations of addition and multiplication. They represent, so to say, functions of these operations.[7]

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*The operator D_x, also written d/dx, is called a differential operator. If applied to a product (uv) the results are given by the formula $D_x(uv) = uD_xv + vD_xu$.

Section E. The new quantum mechanics.

The main problem of the quantum theory is to determine these functions of the operations, so that the solution of certain equations (hamiltonian) may represent the experimental facts. The original equations of the new mechanics of Heisenberg, Born, and Jordan were frankly founded on an empirical basis. As Dirac puts it, in seeking for the new equations, the classical equations were to be retained as far as possible and only the operations by which these quantities are combined were to be altered.

To gain this freedom to alter multiplication the data were first interpreted as matrices. Then it was found, by Born and Wiener, that the matrices could be interpreted as a special kind of operator, which furnished means to calculate the matrices. Carl Eckart independently developed a simple operator calculus for the solution of the quantum problems. In this present work I follow closely the paper of Eckart.[7]

The origin of the new quantum mechanics was an epoch-making paper by Werner Heisenberg, in July, 1925. The older quantum theory had postulated the existence of stationary states of the atom, which were calculated with the aid of the older mechanics. In the new mechanics the equations have similar form as in the classical theory, but the variables no longer obey the commutative law of multiplication. In general pq is not equal to qp, ($pq=qp$ and $pq-qp=0$ in the classical theory) but $pq-qp=\frac{h}{2\pi i}1$, where h represents the Planck constant, q the generalized co-ordinate, p the momentum, 1 stands for the unit matrix, and π and i have the usual meaning. The fact that multiplication is not commutative in our calculus allows us to give a definite value to the above difference and by introducing the Planck constant h, we are enabled to introduce the quantum conditions in our calculations.

The quantum conditions of the older theory led to an algebraic equation. By using the classical equation with a non-commutative multiplication law for the variables it is possible to perform calculations in the new and wider scheme of dynamics. The difference between pq and qp is expressed in terms of the Planck constant h. When h is made to approach zero, pq approaches qp, and so we pass to the classical mechanics. Thus we see that the classical mechanics appear only as a particular case of this more general theory.

In introducing his theory, Heisenberg pointed out that the older mechanics uses quantities which are *never observable*, and *can never be observed*, such as, for instance, orbital frequencies and amplitudes, or the position and 'time' of revolution of an 'electron'. , which, as such, have no physical meaning. He proposes to use *observable* data, such as the frequencies and intensities of the radiations , . Now these frequencies are always differences between two terms given by integers. If T_n and T_m are two such terms, the observable frequency is theoretically represented by $\nu_{nm}=T_n-T_m$. Such numbers as ν_{nm} characterize the atom as far as it is observable. It was natural that such a collection ('sum' in this case has no longer any physical meaning) of terms could best be represented by a matrix. In the classical theory a dynamical quantity was

represented structurally by a trigonometrical Fourier series; in the new, it is represented by a two-dimensional table of values; that is, by a matrix giving the frequencies and the intensities of radiations.

An important and interesting structural issue now appears. It is that the Heisenberg theory gives a new formulation for the hamiltonian equations of motion, whereby their form is preserved, yet they are made applicable both to periodic and to *non-periodic motion*. It becomes possible to fuse the classical mechanics with the quantum mechanics. The distinction between 'quantised' and 'unquantised' motion loses all meaning, and a fundamental equation, $pq - qp = \frac{h}{2\pi i} 1$, is formulated which is valid for *all motion*.

The Heisenberg theory is also characterized by its thoroughly behaviouristic, actional, functional, and operational character. The number of unjustifiable assumptions is the lowest in existence and most of the identifications are eliminated. According to Heisenberg, electrons and atoms do not have the 'same' kind of 'reality' as ordinary *objects* of lower order abstractions. This conclusion, which underlies his whole work, is of particular importance structurally. As we know, differences in character separate different orders of abstraction and since the quantum phenomena belong to a higher order of abstraction they must differ from objects which belong to a lower order of abstraction. In this theory the feelings of 'space' and 'time' are no longer applicable to the 'inside' of the atom—as might be expected.

The distinction between 'inner' and 'outer' electrons in an atom becomes meaningless, since it is impossible to recognize a particular entity among a series of similar entities. In accordance with the new 'space-time' outlook we gain a physical basis for the absolute individuality of some eventual unit.

Because of its structure, the Heisenberg theory is a very fundamental one and there is little doubt that the Heisenberg *methods* will be elaborated further and will be kept as a permanent *checking method* in physics. A theory which is thoroughly behaviouristic, with a minimum of assumptions, will probably remain both a most important instrument of research and an inspiration to physicists and mathematicians.

The Heisenberg theory, again, because of its structure and method, does not lend itself easily to visualization. This is not against the theory. The pictorial representations of lower order abstractions are not to be relied upon. Besides, visualization depends on the *lower centres* and therefore must be represented by a *macroscopic* representation of a continuous (rather than a discrete) character, such as waves , .

If we were to try to describe the Heisenberg theory pictorially, which is obviously difficult to do, we would have to give a *negative* description. We should have to say that what we observe must be considered only as radiations from the location which the atom was supposed to occupy.[8]

There remains but to mention some more characteristics of the Heisenberg theory which seem to have very far-reaching structural and semantic bearings

This theory appears frankly statistical and introduces fundamental probability assumptions. The moment we realize that the human organism is essentially an *abstracting* affair and that the abstracting is performed on different levels, or in different orders, it becomes obvious that statistical methods and probability notions become fundamental.

In the earlier days we used to assume that statistical laws were laws with exceptions. Such an outlook was conditioned by our dealings with macroscopic events. Now, we analyse such macroscopic events in terms of microscopic and sub-microscopic events, the statistical laws become accurate laws, not for individuals but for *groups* of individuals. Because we abstract in different orders, we deal *only* with statistical data, mass effects of different 'packets' of nervous excitement, as is best illustrated by different thresholds in different nervous tissues.

The processes in the higher centres, *more remote from the external world*, deal with a special material, no more with statistical data of 'packets' and averages, but with what we used to call 'inferences', 'inductions'. , which give only the *probability* of happenings. But as we have already seen, probability has become a well-developed structural mathematical discipline, which has not yet made much effect upon our primitive-made macroscopic metaphysics and language. It should be noticed that the highest activities of the nervous centres are based on statistical data furnished by the lower centres. So we see that, to the best of our knowledge about ourselves and the world around us, a modern structural and semantic outlook, in science or in life, must be based on statistical and probability methods.

In space-time every point has a date, and therefore in the language of space-time all points are different and do not repeat themselves. Such structural outlook is, of course, again conditioned, and leads toward the statistical and probability methods. The main psycho-logical importance of the new methods is to diminish affective tension, which is always wasteful and harmful. Inferences may involve belief. When *belief* is *too strong*, although this is never justified according to the best modern knowledge, we very easily fall into identification, delusions, illusions, and the like. It should be emphasized that the last-mentioned pathological states are always compound. They involve at least two components. One of these consists of some ignorance somewhere; the other of strong affective belief in the 'truth' of our mistaken notions. The stronger the affective tension is, the more dangerous the semantic disturbance becomes.

The Heisenberg theory has succeeded in formulating (verbal) structural methods which are best suited to represent the experimental facts which underlie physics, as well as being structurally in accord with the working of the human nervous system. That is why I venture to assert that this theory will never be abandoned as a checking and research instrument.

In an hypothetical experiment in the quantum field, we may assume what may be called a gamma-ray microscope. If we were to illuminate an 'electron'

by gamma-rays, the rays would disturb the experiment, and in our fundamental equation, $pq - qp = \frac{h}{2\pi i}$ 1, by which the 'position' was to be determined, the 'momentum' would thereby be disturbed. This change of 'momentum' would be greater the shorter the wave-length of the rays used; and the shorter the wave-length of the ray, the more accurate the determination of 'position' would be. Hence, the more exactly a co-ordinate q could be found, the less exactly could its momentum p be found, and vice versa.

So we have to introduce corrections for errors, and have to introduce 'mean values' and 'probability functions', which we can develop and compute. Lately, Bohr has further developed the probability aspects of the newer quantum mechanics but I have not seen this work. Heisenberg introduces 'probability packets' which correspond to the 'wave packets' of Schrödinger.

It is difficult to speak briefly, and yet in a satisfactory way, about these new developments, and particularly difficult to give credit properly to different authors. All their works are interwoven and at present they all really work together in spite of the fact that historically some of these theories have been developed independently.

What we call today, for the sake of brevity, the Heisenberg theory, because of its originator, has been further developed by Heisenberg, Born, Jordan, and others. Later, when the wave-mechanics appeared, all the new theories were finally fused into a very elaborate and impressive structure.

Historically, P. A. M. Dirac worked at the theory from a different mathematical point of view, utilizing what is called the 'Poisson bracket' method. In this treatment the difficulties of the matrix calculus were avoided. He introduced dynamical variables which he called the q numbers. These do not obey the commutative law of multiplication although the c numbers (classical) do. He also considered the difference of the non-commutative products $xy - yx$, where x and y are functions, respectively, of the co-ordinates $q_1 \ldots q_s$, and of the momenta $p_1 \ldots p_s$ of a multiple periodic system with s degrees of freedom.

Dirac has generalized the matrix theory and the Schrödinger equations. His work seems to be most important, in physics and mathematics, but it is not possible for us to consider it here in any detail.[9]

Let us recall once more that there are fundamental differences between the different orders of abstraction, and that we all have to abstract in different orders. From this point of view it is natural that *every* theory, even if expressed at present in a form which cannot be *visualized*, like the Heisenberg theory or the original Dirac theory, has sooner or later to be expressed in a structural form which can be visualized. These problems have really nothing, or at most very little, to do with the world around us. They are concerned with the neurological structure which produces *all* theories.

Theories of a structure such as that of the Heisenberg theory are extremely important, as already explained, but in them we lose the help of 'intuition'. Now 'intuition' (lower centres) has two quite different effects—sometimes it leads us astray, but on other occasions it helps greatly.

An 'intuitive' theory has a creative aspect, but always ought later to be revised and scrutinized by non-intuitive means. In fact, because of our nervous structure, we should always strive to produce *both aspects* of theories—strive *consciously*—for thus we facilitate progress. Historically, we can never completely avoid producing both types of theories, as they are inherent in our nervous structure and in the different orders of abstractions we produce.

It is precisely in the newer quantum mechanics that a typical example of this simple neurological fact is found. The non-intuitive handling of data was introduced by Heisenberg: the translation of the matrix calculus into operational and 'Poisson brackets' methods; and, finally, the new 'wave mechanics' of de Broglie, Schrödinger, and others, gives us a perfect translation into intuitive methods.

It should be noticed that according to the old notions such two methods, the intuitive and the non-intuitive, were not supposed to be a *neurological necessity*. We still assumed that they were separated 'absolutely', and even today in many quarters we argue as if they were absolutely separable. If we accept the principle of non-elementalism, we realize that this distinction is verbal only and that the invention of verbal means has little or nothing to do with the world around us, but that it depends on human structural ingenuity.

Investigation of the ordered cyclic nerve currents shows unmistakably that such sharp differentiation is unjustifiable; and we must conclude, in accordance with historical experience, that translation from one method to the other must be a necessity, and so will be accomplished some day in every field. It is true that at present the Einstein theory has not been translated with entire success into terms of lower order abstractions. This is a task which is facilitated by this present work. The newer quantum mechanics gives us an unparalleled example of such translation, and hence our main interest should be concentrated on this structural aspect.

On neurological grounds it seems certain that visualization involves in some way the lower nerve centres, which again, by evolutionary necessity, involve macroscopic forms of representation. Our macroscopic experience led us to *geometrical intuitions*. These were framed three-dimensionally in '*absolute emptiness*', and were impossible in higher dimensions. The old structure represented a static empty 'space' in which nothing could happen and which was thus unfit to represent this world around us where something is going on everywhere.

The new structure represents '*fulness*' or a plenum. We can visualize it as a network of intervals or world-lines and then, by the notions of the differential calculus, as already explained, we pass easily to the visualization of the many-dimensional space-time world of Minkowski-Einstein. Now, in such a world, the curves are represented by functions, and, vice versa, functions represent curves. Thus it is obvious that analytical 'non-intuitive' methods have 'intuitive' structural geometrical counterparts. From this point of view the method of operators represents a passing step from the non-intuitive to intuitive

methods. As soon as we have functions, we can represent a functional calculus as an operational calculus. This involves a more behaviouristic semantic attitude and so leads to the possibility of translation of either method into the other. Ultimately these are psycho-logical transformations and translations.

We should not be surprised to find that in the development of the newer quantum mechanics the operational calculus plays just such a role.

Now our older macroscopic experience, which affected our lower nerve centres, gave rise to the elementary geometrical structural notions of 'lines', 'surfaces', 'volumes', . For the building of physics we had to introduce 'time', 'motion', . In older days we did not realize that these give us forms of representation and that it is optional with us which forms we will accept as fundamental or use as a starting point.

The old descriptive apparatus posited structurally an absolute and immutable 'space' (emptiness), 'time', 'matter' out of which we built up a verbal definition for 'motion'. The semantic attitude of all of us, scientists included, depended upon identification. We ascribed lower-centre significance to higher-centre abstractions. We did not discriminate enough between the macroscopic and small scale events. So we had 'geometrical optics' in which we 'perceived' a ray of light (in a dusty room, let us say) as a 'straight line'. Further investigation disclosed that the 'rays' on one level of abstraction were *waves* on another, but they were not perceived as waves by the lower centres.

But through our lower centres we had acquaintance with some waves, such as in water; so representation for waves was developed. A wave-theory still remained intuitively workable, even when we dealt with waves which we could not see. Now the equations of waves are well known. It is then possible to translate a non-intuitive matrix mechanics, when we treat matrices as operators, into a functional calculus which has an intuitive geometrical wave representation.

This is precisely what has happened and now, perhaps for the first time in human history, we have all the aspects of a theory being worked out simultaneously, with mutual co-operation of all workers and the use of methods which are mutually complementary from the neurological side. There seems little risk in predicting that because of these neurological factors the newer quantum mechanics will give extremely rapid and far-reaching results. When scientists become aware of the structural semantic and neurological issues involved, perhaps such achievements will be multiplied *consciously*, instead of being a kind of coincidence.

Personally, I am convinced that these new achievements are not simply coincident. It seems that the abolishment of the old, *el*, static 'absolute space' and 'absolute time' *has relieved the younger scientists from a semantic blockage*. This release was due to the bold stroke of genius of Einstein in refusing to use the vicious aristotelian 'is' of identity. As soon as we realize that words are not the objective levels, we gain an *unconscious semantic freedom* in handling words, as words. At once this freedom is bound to produce many different forms of

representation for events, according to the personal make-up of the individual workers. And of course these forms can be translated into one another.

Section F. The wave mechanics.

We have not sufficient space at our disposal to discuss more fully the new wave mechanics. I found that short of a small volume, no explanations, readily intelligible, could be made.

In mathematics and physics, which represent the most developed sciences, we consciously and unconsciously strive for more and more general formulations. The work of Einstein, showing that the classical mechanics was only a particular case of a more general mechanics, has given a healthy stimulation to such a fruitful line of work.

As the quantum phenomena could not be accounted for by the old mechanics, it was natural that physicists should try schemes of new mechanics which included classical mechanics as a particular case. Thus, Sommerfeld, through his methods of the application of the Einstein theory to the quantum mechanics and his generalizations of phase-space, and his treatment of the relation of wave-optics to ray-optics and of the relation of mechanics to ray-optics, came close to the discovery of the wave mechanics.[10]

The new wave mechanics originated in 1924 in Paris, with the thesis of Louis de Broglie, published in 1925, and republished in a book form in 1926.

The controversy between the corpuscular light theory of Newton (emission theory) and the wave theory of Huygens is well known. The emission theory had its support in the 'rectilinear' propagation of light, which followed from the inertia of light particles. Also, it explained the reflection and the refraction . , of light, but failed in other respects. It is true that the wave theory also had its weak structural spots. In it the 'rectilinear' propagation of light remained a complete mystery and it completely failed to account for the dispersion of light, until this was explained on the electron theory.

Both theories assumed the periodicity of light phenomena, but the acceptance of one theory was generally held to mean the rejection of the other. It did not occur to many that both theories might be correct but only partial structural aspects of a more general theory.

With the advent of the quantum theory of Planck (1900) new methods were found. In 1905 Einstein propounded his theory of 'light quanta' successfully. He assumed that radiation occurs in discrete quanta of energy $h\nu$, where ν represents the frequency. From this point of view the quantum had the characteristic discreteness of a corpuscle, and yet the frequency characteristic of a wave. We see that the new theory involved a kind of a blend of the two older theories.

De Broglie generalized still further the above notions. His theory is in a way the result of the theory of Einstein. As we already know, Einstein shows the connection of mass and energy, so that the conservation of mass becomes also the conservation of energy, and vice versa. Starting with these premises de Broglie concluded that if any element, in the most general sense, be it an

electron or proton or light quantum or what not, has energy W, there must be in the system a periodic phenomenon of frequency ν, defined by $W=h\nu$. From this point of view all forms of energy, radiation included, must have an atomic structure and the atoms of energy must be grouped around certain points, forming what we call 'electrons', 'light quanta', .

Applying the Lorentz-Einstein transformation, he finds a rather startling fact—that the frequency associated with any assumed mass m_0; namely, $\nu_0=\frac{m_0c^2}{h}$, represents no more and no less than a *periodic* phenomenon, analogous to a stationary wave, which spreads around the point of which the mass is a singularity.[11]

In other words, a 'mass particle' at rest is the centre of a pulsation through-out the spread, or otherwise it is a singularity of the pulsation. The quantity which pulsates is called ψ and $\psi\bar{\psi}$ is interpreted as the *electric density*, where $\bar{\psi}$ is the conjugate of the complex quantity ψ.

In the Minkowski representation, the above astonishing result becomes quite simple, and we can see clearly how simultaneous pulsations become travelling waves. In Fig. 2, we give a two-dimensional diagram of space-time. OX_0 is the 'space' co-ordinate, OT_0 the 'time' (ict_0) co-ordinate. ψ_1, ψ_2, ψ_3, represent the traces of the surfaces of constant phase which are perpendicular to OT_0. The Lorentz-Einstein transformation is equivalent to the transformation from rectangular system X_0OT_0 to the rectangular system XOT, forming an angle θ. In the new system the lines ψ_1, ψ_2, ψ_3 . . . , are no longer parallel to the X axis and so represent a moving *wave front*. In this new system the ψ-lines represent the moving wave front for different points, P, P'. , on one phase line and have *different* values of t. The smaller the velocity (v) of the particle, the smaller the angle θ and the greater the distance PQ travelled by the phase in a given 'time', $P'Q$ or Δt, which means the greater the phase velocity u.

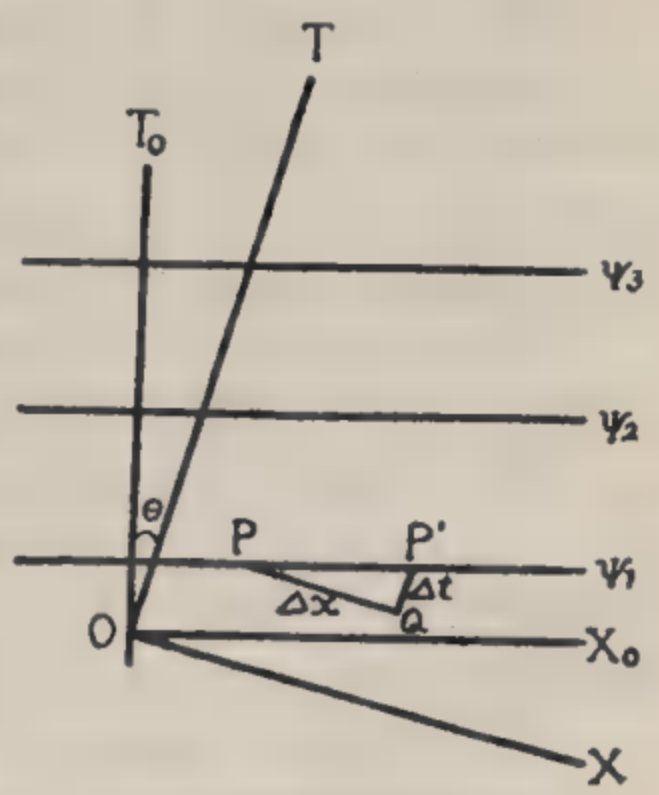

FIG. 2
(From *Wave Mechanics*, by H. F. Biggs)

The frequency of these pulsations or waves is the 'total energy of the particle' divided by Planck's constant, h. In symbols, $h\nu=mc^2+$potential energy, where

$$m=\frac{m_0}{\sqrt{1-\beta^2}},\quad \beta=\frac{v}{c}.$$

The problem before us is to connect structurally, the waves with two observations, one of the radiations, the other of what we call the 'material particles'.

Our interest at present lies only in the connection with the latter. According to this theory, the 'region occupied by the particle' is only a region

where a set of ψ-waves, which vary continuously in direction and in frequency in a small range, reinforce each other to give a *wave-group* travelling with what we usually call 'the velocity of the particle'.

As the waves have different frequencies, they travel with different velocities and so we have to face the problem of the dispersing medium, where, according to classical theories, the region of reinforcement has a velocity different from that of the phase. The ordinary expression for 'group velocity' gives, on the wave interpretation, the magnitude of the 'velocity of the particle'.[12]

That which, in the classical theories, we called the 'motion of the particle' is represented by the motion of the region of reinforcement of ψ-waves. The direction of motion is represented by the direction of a ray-, or wave-normal. The particular ray selected as the position of the 'path of the particle' is represented by the ray cut in coinciding phase by a set of ψ-waves with slightly varying directions. The 'position of the particle' is given by the small region occupied by the *group* of waves of slightly different frequencies and velocities.

In this connection we should notice an important point which is necessitated by the methods of generalization and of translation from macroscopic to sub-microscopic events and vice versa. Schrödinger, by formulating the differential equations for the ψ-waves, has brought out clearly the important structural point that wave mechanics bears a similar relationship to classical mechanics of particles as wave-optics bears to ray-optics.

Here again we have the macroscopic phenomena capable of being treated by mathematical methods different from those used for the small-scale phenomena.

In classical mechanics the state of a system whose co-ordinates were $q_1 \ldots q_s$ and whose momenta were $p_1 \ldots p_s$, was represented by a point in a $2s$-dimensional q-spread and the changes in the system were represented by the passage of the point along some curve, a 'ray', so to say.

Schrödinger regards the classical mechanics as only an approximation, while rigorous treatment must be made by the aid of wave mechanics.

The large-scale, or macroscopic, mechanical processes correspond to a wave signal in the q-spread and can be regarded as a point in comparison with the geometrical structure of the path. In small-scale phenomena, such as the atomic processes, a rigorous wave formulation must be used.

This analysis can be carried further and Hamilton knew well and used the analogy between mechanics and geometrical optics. The Hamilton variation principle, $\delta \int L dt = 0$, is Fermat's principle for a wave motion; the Hamilton-Jacobi equation expresses Huygens' principle for wave motion; and the new wave mechanics expresses the Kirchhoff analysis of physical optics. As Huygens' principle could deal with the problems of physical optics up to a certain point, so the Hamilton-Jacobi equations could deal with atomic problems up to a certain point. At the exact wave analysis of Kirchhoff was needed to clear up the finer points of physical optics, so the new wave mechanics is required for the exact solution of the atomic problems.[13]

A detailed analysis shows that classical mechanics was associated with geometrical optics (ray-optics). Obviously, a more exact system of mechanics

would be one associated with wave-optics, which would give the classical results in all cases where the wave-length was negligible in comparison with the dimensions of the path. Schrödinger suggests that a correct extension of the analogy would be to regard the wave system as *sine-waves*. In this connection it should be recollected that Fourier has shown that any given form of waves can be represented by the superposition of *sine-waves*, and that therefore a *sine-wave* (see Chapter XXXII) may be considered as a general formulation.

The ray methods in physics worked only to some extent, that is to say, in the cases where the radii of curvature of such rays and the dimensions of the spreads were large in comparison with the wave-length. When this is not the case we have to consider waves and not rays. Naturally, dealing with atomic dimensions, which are very small, instead of using the paths of the particles or the ψ-rays, we have to use the ψ-waves. It appears that this difference was the main, rather puzzling distinction between the classical mechanics and the quantum mechanics, between the macroscopic theories and the sub-microscopic ones.

The above realization, and its formulation into a mathematical theory, seems to be an important and extremely fruitful generalization, which probably will be retained as a method.

One of the puzzling features of the quantum theory was the structural appearance of the whole-number laws of the 'orbits'. That some such whole-number relation is justified seems to be well established, yet it contradicted the older 'continuous' mechanics. A new theory, to be at all satisfactory, should be able to fit these whole-number empirical data. The first test of the new wave mechanics, and also its first success, was precisely in this field.

If a ray of the ψ-wave was supposed to run around in a circle for a stationary state, the circumference must be a *whole multiple* of the wave-length, or

$$2\pi r = n\lambda = n\frac{h}{mv}, \text{ and } mvr = n\frac{h}{2\pi}$$

where n is an integer. We see that the quantum condition of the Bohr theory, that the angular momentum must be a whole multiple of $h/2\pi$, is only the result of the requirement that the wave-function ψ shall be single-valued, which is another way of saying that the circumference ($2\pi r$) must contain a whole number of wave-lengths. It may be compared perhaps with waves travelling around a circular loop of string. If they travelled both ways we would have stationary waves.

At this point a very important structural feature of the new wave mechanics makes its appearance. In the above interpretation the 'velocity of the electron' has lost its physical meaning, it becomes simply the *wave length* of the ψ-waves. In the wave mechanics, as well as in the matrix mechanics, there is no meaning to the older 'position of an electron on its orbit'. So the wave mechanics again embodies the advantages of the matrix mechanics by not postulating entities which can never be observed. The whole numbers, as Schrödinger remarks, 'appear as naturally as do "integers" in the theory of vibrating strings'. In the theory of string vibrations these whole numbers are determined by certain

boundary conditions which have to be satisfied by the solution of a differential equation. In the new wave mechanics there is also a differential equation representing the Schrödinger wave equation.[14]

Section G. Structural aspects of the new theories.

We should notice the important distinction between the two structural types of these theories. The extensional matrix theory can hardly be visualized, with all the consequent advantages and disadvantages. The wave mechanics can be visualized. From what we already know of the structure and working of the nervous system, we see that the wave mechanics will have a *creative* element and the matrix mechanics will remain an important *checking* method.

At present, all these new theories seemingly have blended or perhaps it would be better to say that they have been translated from one language to another and all the workers in this field work from all angles.

It should be mentioned also, that Einstein, Bose, Jordan, and others, work from the point of view of *statistics*, and that these methods, too, are being retranslated and connected with the rest of the new theories.

The new wave mechanics evades the difficulties of the matrix calculus and brings the new mechanics within the scope of the highly developed analysis of the theory of differential equations. It also enlists the creative aspects of 'intuition', 'visualization', .

Concluding our consideration of the subject, three remarkable aspects of the wave mechanics must be referred to. We are already acquainted with the term 'action'. It appears that the main point of the passing from the old mechanics to the new was the stroke of genius of de Broglie, when he divided action by the fundamental constant h with some definite numerical factor which then gives us the *phase*. In the expression for ψ, the energy appears as the 'time' component of a space-time vector whose 'space' components are those of the momentum. When this vector is divided by h, its components become the *frequency*, or the number of waves which each axis cuts per centimetre.

These are the methods by which we can use differential equations, whereby the older discontinuities disappear and the particle is represented as a group of reinforcing waves.[15]

From this point of view we also come to the conclusion that the 'conservation of energy', which was very valuable in the old days, is perhaps only a gross macroscopic generalization and will give place to a newer and more fundamental notion of the conservation of *frequency* or 'times'.[16]

It has been already mentioned that the newer mechanics must be represented in accordance with *statistical* data, *probabilities*, with due attention paid to the theory of errors , . While these requirements have very little to do with the world around us, they are unconditionally required by our nervous structure, which is, after all, the general author of all our 'knowledge' and 'theories'. Let us be candid about it; there is no such thing as 'knowledge' outside of a nervous system, and therefore the neurological requirements, as already

mentioned, become paramount. The newer theories brilliantly satisfy this requirement.

As an example, we may perhaps mention an aspect of the wave mechanics theory which is not at present settled, but which remains just as interesting.

Schrödinger shows that in highly excited states a suitably chosen group of waves represents a '*wave packet*', which behaves like a point-mass of the ordinary mechanics. It oscillates with frequency ν_0 in a rectilinear path. The number and breadth of the waves which form the packet vary with the 'time', but the width of the packet remains constant. The remarkable part about the shape of the curve is that it represents the *Gauss error curve.*

Heisenberg has shown that this result is only accidentally true, but for our purpose of illustration this is quite enough.

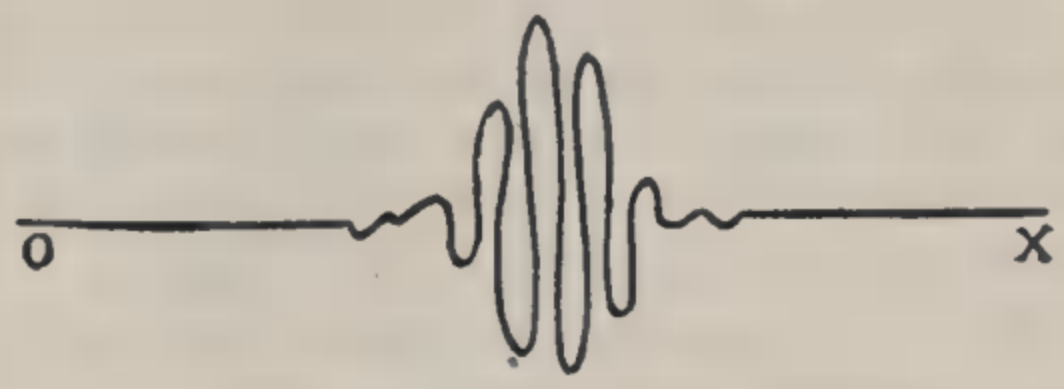

FIG. 3

(From *The New Quantum Mechanics*, p. 115, by George Birtwistle)

The newer quantum mechanics has shown once more the necessity for a re-analysis of our fundamental notions. From a space-time point of view, which seems to be a permanent acquisition of science, since it is a language and method of structure closer related to the external world than the older languages and methods, it seems beyond dispute that even macroscopic phenomena are the results of *repeated observation.* Now, such a point of view, although it is extremely plausible, and close to neurological and physical data, necessitates a complete reconstruction of our describing apparatus, which is not adapted to such an outlook.

The problem of 'observation' enters. Bohr suggests, and rightly, that this vocabularly is strictly connected with the older 'causal' vocabulary. One of the main points of the present work is to draw attention to the *multiordinal mechanism and terms* and to show that the analysis of these problems cannot be even attempted without first analysing the structure of our languages, of our 'knowledge', and the neurological and semantic aspects which such analysis involves.

When this analysis is carried through we see that the problems of 'continuity' and 'discontinuity' lose their absolute character. They become *verbal* problems, to be solved through the ingenuity of some one who will suggest the solutions.

The newer quantum mechanics give us ample material for work on these problems, but they also illustrate a much more general and important problem, which is the subject matter of the present book; namely, that all 'knowledge' is *structural*, strictly dependent on the nervous structure and functioning, and the language we use. 'Method' is that aspect of the search for structure which deals with the most expedient means for finding structure. Since words *are not* the things we speak about, the study of linguistic structure becomes a

most important research method. The more languages (theories) we have for analysis and structural comparison, the more glimpses do we get at the structure of the world. The newer quantum mechanics give us enormous material of a linguistic, structural, and semantic character.

It is natural that this wisdom can come only from the study of the structure of the highest developed languages in existence, which are mathematical languages. If we want progress in any line of human endeavour, this progress is always dependent on the languages we use, since what we call 'progress' is a co-operative affair and therefore dependent on means of communications and languages.

From the point of view of structure, we deal with a world of *absolute individuals* and therefore our languages must be such as to reflect such individuality. We already know that this involves an extensional attitude and methods, which historically have produced mathematics as the only language which as yet reflects in structure the world around us.

With the newer quantum mechanics, the old 'discontinuity' resolves itself into an essential *individuality*, as noticed by Bohr, perfectly foreign to the older theories.

History proves that we were slow in arriving at that point. Our tragedies began when the 'intensional' biologist Aristotle took the lead over the 'extensional' mathematical philosopher Plato, and formulated all the primitive identifications, subject-predicativism . , into an imposing system, which for more than two thousand years we were not allowed to revise under penalty of persecution. Mathematics was not particularly encouraged, but at least, not persecuted, so that it was developed into the present day great linguistic system. The theory of function involves semantic factors of non-identification.

The invention of the differential and the integral calculi, represents the two great structural and psycho-logical aspects of analysis and differentiation, versus synthesis and integration.

The application of these methods led us to differential geometries, to methods of treating 'fulness', and to 'contact' methods. 'Fulness' necessitated geometries of higher dimensions, impossible in 'absolute emptiness', and so the fusion of geometry with physics became possible. The four-dimensional world of Minkowski and the theory of Einstein finally achieved this fusion. The next step was the invention of the new quantum mechanics, where all these important, nay, all-important, structural, semantic, and linguistic achievements find their culmination. The old primitive metaphysics become too 'materialistic' for an enlightened age.

Without legislating about the 'truth' or 'falsehood' of the newer mechanics, as a matter of *human behaviour* these theories are the best indications and examples of the structure of human 'knowledge', which I have attempted to formulate in this work as a general theory.

The A-system was strictly interconnected with primitive-made structural assumptions or metaphysics, reflected in the structure of the older languages and in the *el* notions about language, 'psychology', 'logic', and the pre-scientific

anthropomorphic astronomy, physics, and other disciplines. Of late, science has developed in spite of all handicaps and persecutions, and has begun to depart structurally and semantically from the path of aristotelianism and the dark ages. Every science has had to build its own language and this fact completely condemns the *A* language, which, it is shocking to notice, we continue to preserve in our daily life.

Should we wonder that we have shown hardly any progress at all in our purely human affairs and notions? We should wonder, rather, that we have been able to survive until now—though with needless difficulties and suffering. More wars, more revolutions, more insanity, more morons, more struggle and competition, and more unhappiness are what we are entitled to *expect and predict* as the outcome of this structurally and semantically impossible situation.

As the organism works as-a-whole, such things as 'pure intellect' or 'pure emotions' represent structurally *el* fictions and scientists should realize that their professed detached scientific attitude is profoundly and fundamentally unjustified. All science has 'emotional' components, which play most important roles in life. If we live in a modern world, but keep the 'emotional attitudes' of primitive bygone days, then naturally we are bound to be semantically unbalanced, and cannot be adjusted to a fundamentally primitive 'civilization' in the midst of great technical achievements.* When scientists understand that, then the layman will have a different attitude toward science. He will understand that science is not a privilege of the few, something without effect upon all and every one. He will realize that while he lives in a modern world, *made so* by science, *structural ignorance* of the fundamentals as discovered by science leaves him with primitive structural assumptions or metaphysics, which by necessity build for him a delusional world leading to semantic unbalance and ultimately to 'mental' and nervous ills.

From the *non-elementalistic* point of view, the only escape is to realize that ignorance in an adult is, and must be, pathological, because 'knowledge' is to be considered as a normal characteristic of human *nervous tissue*.

A special structural and methodological brief and simplified account of scientific achievements, such as I have attempted in this work, must be a part of a theory of *sanity*. Sanity means *adjustment* and without the minimum of the best structural knowledge of each date concerning this world, such adjustment is impossible.

It is not necessary that the reader should fully understand all technical details of a theory, to be *aware* (instinctive, affective . ,) of the *existence* of the structural and semantic problems and to realize that some of the most competent and skilled professionals are working at these theories. Such awareness has great pacifying semantic influence; it eliminates the older affective tensions which were due to identifications, absolutism, dogmatism, flights into mysticism, and other similar pathological disturbances.

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*See my *Manhood of Humanity*.

A *non-aristotelian system* must deal with all these structural and semantic issues. It is hoped that once a $\bar{A}$-system is presented to the public, scientists and laymen will become more interested in the structural and semantic issues emphasized here, and that new and wider researches will be undertaken.

History shows that such hopes are not illusory. The greatest men of science have always had wide human aims and interests. From the *non-elementalistic* point of view, they probably became productive geniuses because of this broad human urge. From the point of view of psychiatry, it is well known that 'mental' ills involve usually anti-social affective attitudes. When we see men with distinctly anti-social tendencies, no matter how they rationalize them, they are invariably ill in some way. A fully healthy individual is never anti-social.

That science should include structural and semantic factors of sanity may be a startling notion, but only at first! In the present analysis this turns out to be, rather unexpectedly, a necessity. But on second consideration we should rather expect it. Science and mathematics show the working of the 'human mind' at its best. Accordingly, we can learn from science and mathematics how this 'human mind' should work, *to be at its best.* Then we should make an analysis of science and mathematics from some wider structural and semantic point of view,—the task which has been undertaken in the present work.

At this early stage it is, of course, of comparatively little importance to what extent this analysis turns out to be satisfactory. The main point is that it has been *originated.* If the present author fails, others, perhaps even because of his failure, may be stimulated to do it better. The great and vital thing is that it should be done, by someone.

SUPPLEMENT I

THE LOGIC OF RELATIVITY

BY R. D. CARMICHAEL

In order to be able to deal with such quantities as are involved in the measurement of motion, time, velocity, etc., or indeed in the quantitative analysis of any physical phenomena, it is necessary to have some system or systems of reference with respect to which measurements can be made. Let us consider any set of things consisting of objects and any kind of physical quantities whatever, as electric charges or magnets or light-sources or telescopes or other objects and instruments, each of which is at rest with respect to each of the others. Let us suppose that among the objects are clocks, to be used for measuring time, and rods or rules to be used for measuring length, and that time and length may be measured at any desired instant and any assigned place. Such a set of objects and quantities and instruments, including the equipment for measuring time and length, all being at rest relatively to each other, we shall call a system of reference. Such a system we shall denote by S. In case we have to deal at once with two or more systems of reference we shall denote them by S, S', S_1, S_2, . . .

In this definition of systems of reference nothing specific has been said about the units of length and of time. If we were dealing with our usual principles of mechanics we might pass over such a matter without any feeling of difficulty about it; it would be sufficient to proceed in accordance with our intuitive conceptions of time and length. But in the theory of relativity these appear in a new light. We can not proceed with confident dependence upon our intuition. On the other hand we shall not attempt to give explicit definitions of units of time and length. We shall proceed from certain principles or postulates, presently to be stated, to an analysis of time and length and so arrive at a suitable precision of these conceptions by means of certain guiding principles. It will be seen that it is not far from the truth to say that our fundamental terms are defined implicitly and indirectly by means of the statements made about them and accepted initially as valid and that they may mean anything which is consistent with the truth of these fundamental principles and postulates.

The restricted principle of relativity may now be stated in the following form:

RESTRICTED PRINCIPLE OF RELATIVITY. *If S_1 and S_2 are two systems of reference having with respect to each other a uniform unaccelerated motion, then natural phenomena run their course with respect to S_2 in accordance with precisely the same general laws as with respect to S_1.*

This principle says nothing about the suitability of any particular system of reference for the convenient expression of the laws of nature; but it does say

that if either S_1 or S_2 is suitable the other is equally suitable, the relative motion of the two being unaccelerated.

In order to bring into suitable relations the measurements made on one system of reference and those made on another it is necessary to have some agreement as to the correspondence of units on the two systems. Accordingly we shall make the following assumption concerning the correspondence of units:

PRINCIPLE OF CORRESPONDENCE OF UNITS. *The units of any two systems S_1 and S_2 are such that the same numerical result will be obtained in measuring with the units of S_1 a quantity L_1 and with the units of S_2 a quantity L_2 when the relation of L_1 to S_1 is precisely the same as that of L_2 to S_2.*

We shall agree that the restricted principle of relativity is to be understood in a sense which implies this assumption concerning the correspondence of units; that is, the latter will be taken as a more precise formulation of a part of the content of the former. It is clear that the possibility of realizing this latter is taken for granted in the Galileo-Newtonian mechanics; it is often passed over without remark although it is a profound fact and is a part of the essential basis of any theory of motion.

It is a grave question whether the restricted principle of relativity can be maintained in the interpretation of natural phenomena. Indeed in the more general theory of relativity, to be taken up later, it is treated merely as a sort of approximation to a more comprehensive principle—an approximation strictly valid only in the absence of a gravitational field but very close to the truth for a wide variety of phenomena including most of those which are purely terrestrial.

There are two particular characteristic postulates, or 'laws of nature', lying at the base of the restricted theory of relativity. These may be stated as follows:

POSTULATE M. *The unaccelerated motion of a system of reference S can not be detected by observations made on S alone, the units of measurement being those belonging to S.*

POSTULATE R. *The velocity of light, in free space, measured on an unaccelerated system of reference S by means of units belonging to S, is independent of the velocity of S and of the unaccelerated velocity of the light-source.*

For these two particular postulates there is the strongest possible experimental evidence. Everything known points toward their truth, and there is nothing known which in any way seems to be in disagreement with them. It is to be observed that they apply only to the ideal case, that is, the case in which there is supposed to be no gravitational field.

For the development of the restricted theory of relativity there are three additional necessary postulates, or 'laws of nature;' those that theory shares in common with the Galileo-Newtonian mechanics. Such assumptions in some form are essential to the initial arguments and to the conclusions which are drawn by means of them. To the present writer it seems to be preferable to have these assumptions explicitly stated. They may be put into the following form:

POSTULATE V. *If the velocity of a system of reference S_2 relative to a system of reference S_1 is measured by means of the units belonging to S_1 and if the velocity of S_1 relative to S_2 is measured by means of the units belonging to S_2 the two results will agree in numerical value.*

POSTULATE T. *If two systems of reference S_1 and S_2 move with unaccelerated relative velocity and if a body moves relatively to one of the systems in a straight line with unaccelerated velocity then it also moves in a straight line relatively to the other and with unaccelerated velocity.*

POSTULATE L. *If two systems of reference S_1 and S_2 move with unaccelerated relative velocity and if a line segment l is perpendicular to the line of relative motion of S_1 and S_2 and is fixed to one of these systems, then the length of l measured by means of the units belonging to S_1 will be the same as its length measured by means of the units belonging to S_2.*

We now have before us the logical basis upon which may be built the restricted theory of relativity in all its details. It has been put in essentially the same form as that employed in my "Theory of Relativity" (published by Wiley and Sons, New York) and in my earlier articles in "The Physical Review". Reference may be made to the book named for the detailed development of the theory. Here we shall attempt to sketch only the progress of ideas and to indicate the main conclusions.

The first thing to be done in developing the theory on this basis is to consider carefully the relation between the time units of the two systems. The following remarkable conclusion is reached by a process of reasoning which is fully cogent in character:

If two systems of reference S_1 and S_2 move with a relative velocity v and β is the ratio v/c of v to the velocity c of light as measured on either system, then to an observer on S_1 the time unit of S_1 appears to be in the ratio $\sqrt{1-\beta^2}:1$ to that which is described to him as a unit by an observer on S_2 while to an observer on S_2 the time unit of S_2 appears to be in the ratio $\sqrt{1-\beta^2}:1$ to that which is described to him as a unit by the observer on S_1.

Thus we have the extraordinary conclusion that the time units of the two systems of reference S_1 and S_2, not at rest relatively to each other, are of different lengths in such a way that an observer on either system thinks that the time unit of the other system is greater than his own. It is evident that no simple change of the unit on either system (or both) will bring the units into agreement for observers on both systems. As postulates V and L and T are generally accepted and have not elsewhere led to such strange conclusions it is natural to suppose that the strangeness here is not due to them. In the argument the restricted principle of relativity needs to be used only in so far as it is involved in the conclusion that the units of any two systems of reference S_1 and S_2 are such that the same numerical result is obtained in measuring with the units of S_1 a quantity L_1 and with the units of S_2 a quantity L_2 when the relation of L_1 to S_1 is precisely the same as the relation of L_2 to S_2. But this principle is accepted in the classical mechanics and has not elsewhere led to strange results. The conclusion in postulate M appears to be demanded

by the strongest experimental evidence; it is generally accepted; if the strange element in the result concerning units of time is due to this postulate, it appears that we must accept it as being required by such experience as has already been tested with due care. Hence the conclusion seems to be inevitable that the strangeness in our result is due principally to postulate R.

We shall presently see that the same basis of postulates leads to the conclusion that corresponding units of length in the two systems are also different when taken in certain directions. From the transformations of time and space which result from the conclusions thus obtained the whole restricted theory of relativity may be deduced (as is shown in the book mentioned). Therefore this theory depends essentially on the principle of correspondence of units in two systems of reference and on the propositions set forth explicitly in the postulates; and all of these are either generalizations from experiment or statements of laws which have usually been accepted. Hence we conclude: *The restricted theory of relativity may be developed by logical processes from the generalized results of certain experiments together with certain laws which have for a long time been accepted.*

The main result concerning the relation of units of length may be put in the following form:

If two systems of reference S_1 and S_2 move with a relative velocity v and if β is the ratio v/c of v to the velocity c of light as measured on either system, then to an observer on S_1 the unit of length of S_1 along the line of relative motion appears to be in the ratio $\sqrt{1-\beta^2}:1$ to that which is described to him as a unit by an observer on S_2 while to an observer on S_2 the unit of length of S_2 along the line of relative motion appears to be in the ratio $\sqrt{1-\beta^2}:1$ to that which is described to him as a unit by the observer on S_1.

These remarkable conclusions concerning units of length in two systems of reference rest on just those postulates which led to the strange results as to the units of time.

What often impresses one as the most remarkable conclusion in the theory of relativity is one which implies that the notion of simultaneity of events happening at different places is indefinite in meaning until some convention is adopted as to how simultaneity is to be determined. In fact, *there is no such thing as absolute simultaneity of events happening at different places.* With respect to the measured time and space of physics we must conclude that time does not run its course independently of space. Measured time and space are indissolubly bound together. The theorem which sets this forth most concretely may be stated in the following way:

Let two systems of reference S_1 and S_2 have an unaccelerated relative velocity v. Let an observer on S_2 place two clocks in the line of relative motion of S_1 and S_2 and adjust them so that they appear to him to mark simultaneously the same time. Then to an observer on S_1 the clock on S_2 which is forward in point of motion appears to be behind in point of time by the amount

$$\frac{v}{c^2} \cdot \frac{d}{\sqrt{1-\beta^2}},$$

where c is the velocity of light, $\beta = v/c$, and d is the distance between the two clocks as measured by the observer on S_1.

By means of the foregoing theorems we may readily obtain the formulae for the celebrated Lorentz transformation of space and time coordinates. (The non-mathematical reader may omit the remainder of this paragraph.) Let two systems of reference S and S' have the relative velocity v in the line l. Let systems of rectangular coordinates be attached to the systems of reference S and S' in such a way that the x-axis of each system is in the line l and that the two x-axes have the same positive direction, and let the y-axis and the z-axis of one system be parallel to the y-axis and z-axis respectively of the other system and have their positive senses in the same directions. Let these two systems of axes coincide at the time zero. Furthermore, for the sake of distinction, denote the space and time coordinates on S by x, y, z, t, and those on S' by x', y', z', t'. Let us suppose that S' moves with respect to S in the direction of increasing values of x. Then it turns out that the foregoing theorems imply the following relations between the two systems of coordinates:

$$t' = \frac{1}{\sqrt{1-\beta^2}} \left(t - \frac{v}{c^2}x\right),$$

$$x' = \frac{1}{\sqrt{1-\beta^2}} (x - vt),$$

$$y' = y,$$

$$z' = z,$$

where $\beta = v/c$ and c is the velocity of light.

The foregoing theorems, or (in more compact language) the foregoing equations of transformation, furnish the effective means for developing the whole of the restricted theory of relativity. Our purpose does not require us to follow that development further in detail. But we may mention a few of the remarkable conclusions which now emerge readily. If two velocities, each of which is less than c, are combined the resultant velocity is also less than c. The mass of a body increases with an increase in its velocity relative to the system on which the mass is measured. The mass of a body at rest appears to be the measure of its internal energy. Mass and energy in general appear to be essentially convertible terms. The velocity of light is a maximum which the velocity of a material body may approach but can never equal or exceed.

The development by Einstein in 1905 of the foregoing restricted theory of relativity led to a fresh analysis of the whole foundations of physics. This was made inevitable by its effective attack upon such fundamental notions as those of length and time and mass and velocity. Einstein himself succeeded in 1915 in greatly extending the range of his theory, developing what has since

been called the general theory of relativity. We shall now speak briefly concerning the foundations underlying the latter.

Already in the restricted theory time and space had become essentially blended so that we could no longer speak of a three-dimensional space as separate and apart from the one dimension of time. A sort of combination of the two came into our conception and we began to realize that they can not be disentangled by the measurements of physics. We are forced to consider a four-dimensional continuum of space and time. It is with this space-time extension of four-dimensions that the general theory of relativity has essentially to do; and its problems are intimately connected with the relations of two systems of reference of the generalized sort which this makes necessary. The Lorentz transformation was a great psychological (and even logical) aid in the formation of the new theory.

Let us consider a four-dimensional extension in which space and time are intimately connected and blended so that each point P in these four dimensions represents a definite place A at a definite time t at which A is to be considered. In the course of time a material particle is represented by a succession of these points P. All these points for a given material particle lie on what is called the "world-line" of that particle; and this world-line represents the state of motion (or eventually the state of rest) of the material particle. If two objects come into coincidence at an instant their world-lines have a corresponding intersection. The things which the physicist deals with ultimately are these intersections of world-lines.

In order to deal with them he finds it necessary to introduce certain reference numbers which we may call the coordinates x_1, x_2, x_3, x_4. These numbers change in such a way that their variation along any world-line is continuous and that no two points ever have the same ordered set of four numbers assigned to them. This gives us a very general set of coordinates. It is clear that coordinates can be set up in an immense variety of ways so as to have these few very general properties. One of the first problems in the general theory of relativity is that of the character of the transformation by means of which we can pass from a given choice x_1, x_2, x_3, x_4 of coordinates to a second one ξ_1, ξ_2, ξ_3, ξ_4. It is clear that we must have relations of the form

$$\xi_i = f_i\,(x_1, x_2, x_3, x_4),\ i = 1, 2, 3, 4,$$

where the functions f_i of the variables x_1, x_2, x_3, x_4 are rather general functions of these four arguments and are indeed to a large extent arbitrary. Now suppose that the laws of nature are expressed in terms of the coordinates x and also in terms of the coordinates ξ; the question arises as to what relation one ought to expect between these two forms of the law. Now there are no coordinates in nature. These have been inserted by us for our convenience. What is more natural, then, than the demand that we shall formulate our statements of these laws so that they shall have the same form in these two systems of reference, and indeed in all possible systems of reference? This is precisely one of the fundamental basic requirements upon which Einstein insists. The corresponding

principle he has called the principle of covariance. In detailed and precise form, it may be stated somewhat as follows:

PRINCIPLE OF COVARIANCE. *The laws of nature can be (and are to be) expressed in such mathematical form in terms of the space-and-time coordinates* x_1, x_2, x_3, x_4 *that they shall remain invariant under every transformation of the form*

$$\xi_i = f_i(x_1, x_2, x_3, x_4),\ i = 1, 2, 3, 4,$$

where the functions f are subject to the following conditions:

1) *They are (apart from exceptional points or regions of fewer than four dimensions) finite and continuous and indefinitely differentiable;*

2) *They are such that the transformation is uniquely reversible, the inverse transformation having the properties demanded for the direct transformation;*

3) *They are such that in both the transformation and its inverse the fourth variable has the character of a time variable while the other three have the character of space variables.*

This principle demands the attainment of an ideal which is admittedly mathematical in its character. By means of it alone one could not come to grips with phenomena. One needs some additional hypotheses. One of these is to the effect that the restricted theory of relativity is valid in free space, that is, in space free of a gravitational field. The other is the celebrated law of the equivalence of gravitational forces and the apparent forces due to acceleration. This may be set forth as follows:

PRINCIPLE OF EQUIVALENCE. *For an indefinitely small region of the world (that is, a region so small that the variation of gravitation in it in both time and space is negligible) there exists a coordinate system* S_0 (X_1, X_2, X_3, X_4) *with respect to which gravitation has no influence either upon the motions of mass particles or upon any other physical phenomena whatsoever.*

Such is the logical basis from which the general theory of relativity proceeds. We can not here follow it in its high enterprise of conquest over the laws of nature. The road (at present and perhaps for a long time to come) can be followed only by one who is willing to give serious and long-continued attention to the study of certain branches of mathematics. In the earlier parts of the argument the reasoning is rather technical and abstruse in character and the general steps are intelligible only to those who have a considerable acquaintance with a certain range of mathematical ideas. After a time the exposition comes down, if not to earth, at least to the solar system and cases begin to appear in which it is possible to find means for choosing between the theory of Newton and that of Einstein.

Three crucial phenomena have been brought to light by means of which to test between the two theories. We shall now speak briefly of each of these.

For a long time astronomers have known that there is a certain forward advance in the perihelion position of the planet Mercury which can not be accounted for on Newton's theory. It amounts to about 42 seconds of angular measure per century. This is well accounted for by Einstein's theory.

Einstein predicted, on the basis of his theory, that a ray of light from a star which is seen apparently close to the edge of the sun would be found to be

bent out of a straight path and that the deflection thus caused would turn out to be 1.74 seconds of angular measure, the bending being in such a direction that the star could actually be seen when just behind the edge of the sun. The prediction has been verified with a good degree of precision, observations having been taken at two eclipses of the sun.

A third crucial phenomenon is associated with the vibrations of an atom in a gravitational field. Since the periods of an atom furnish a sort of natural clock, it should give an invariant measure of an interval of time. Proceeding from this hypothesis one concludes that an atom vibrates more slowly on the sun than on the earth, due to the influence of the larger gravitational field of the sun. Hence the lines of the spectrum should be displaced towards the red. For the part of the spectrum usually observed this amounts to about .008 tenth-meters (a tenth-meter $=10^{-10}$ meters). For a long time there was grave doubt whether this phenomenon is actually existent; but the evidence for its existence now (1933) seems to be conclusive.

Moreover in recent years it has come to be recognized that the stars known as white dwarfs have masses which are comparable with that of the sun, while their radii are much smaller. The companion of Sirius is a star whose radius is about 1/35 of that of our sun. Computation shows that the shift in the lines of the spectrum produced by light passing near this star should be about .30 tenth-meters. This matter was put to the test at Mount Wilson Observatory and an actual shift of .32 tenth-meters was found. One would conclude then that it is now hardly possible to doubt the actual existence of the spectral shift predicted by the Einstein theory.

Whatever may be the final verdict concerning the validity of the theory of relativity as a whole, it has certainly made a fundamental and permanent contribution to astronomy in developing a modification of Newton's law of gravitation. It has been checked experimentally in three very different ways and is thus established on a rather secure basis. Three such conquests as those just recorded have probably never before been made so nearly simultaneously by a single theory developed from one point of view consistently maintained throughout.

SUPPLEMENT II

THE THEORY OF TYPES[1]

BY PAUL WEISS*

It would seem from the interpretation that Whitehead and Russell put on the theory of types, that it is impossible or meaningless to state propositions which have an unrestricted possible range of values, or which, in any sense, are arguments to themselves. Thus on the acceptance of the principle that statements about all propositions are meaningless,[2] it would be illegitimate to say, "all propositions are representable by symbols," "all propositions involve judgment," "all propositions are elementary or not elementary," and if no statement could be made about all the members of a set,[3] it would be impossible to say, "all meanings are limited by a context," "all ideas are psychologically conditioned," "all significant assertions have grammatical structures," etc., all of which are intended to apply to themselves as well. The theory seems also to make ineffective a familiar form of refutation. General propositions are frequently denied because their enunciation or acknowledgment depends on the tacit supposition of the truth of a contradictory or contrary proposition. Such refutations assume that the general proposition should be capable of being an argument of the same type and to the same function as its own arguments, so that according to Whitehead and Russell, they fallaciously refute "by an argument which involves a vicious circle fallacy".[4]

That these limitations on the scope of assertions or on the validity of refutations are rarely heeded is apparent even from a cursory examination of philosophical writings since 1910. Thus Russell, apropos to Bergson's attempt to state a formula for the comic says,[5] "it would seem to be impossible to find any such formula as M. Bergson seeks. Every formula treats what is living as if it were mechanical, and is therefore by his own rules a fitting object of laughter." The characterisation of all formulæ, even though it refers to a totality, seems to Mr. Russell to be of the same type as the formulæ characterised.

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[1] Chap. II., *Principia Mathematica.*

*[*Reprinted from* MIND: *a Quarterly Review of Psychology and Philosophy. Vol. XXXVII., N.S., No.* 147; with minor corrections.]

[2] P. 37, *ibid.* (second edition). [3] P. 37, *ibid.* [4] P. 38, *ibid.*

[5] "Prof. Guide to Laughter," *Cambridge Review,* Vol. 32, 1912, and Jourdain's *Philosophy of Mr. B*tr*nd R*ss*ll,* pp. 86-7.

If the theory were without any embarrassments of its own, and were indispensable for the resolution of the so-called paradoxes[1] (which no one seems to believe), there would be nothing to do but to acknowledge the impossibility of cosmic formulations, as well as the inadequacy of philosophic criticisms, and to pass charitably over such remarks as Russell's as mere accidents in a busy life. However, the statement of the theory itself involves the following difficulties in connection with (1) its scope, (2) its applicability to propositions made about it, and (3) its description.

1. It is either about all propositions or it is not.

A. If it were about all propositions it would violate the theory of types and be meaningless or self-contradictory.

B. If it were not about all propositions, it would not be universally applicable. To state it, its limitations of application would have to be specified. One cannot say that there is a different theory of types for each order of the hierarchy, for the proposition about the hierarchy introduces the difficulty over again.

2. Propositions about the theory of types (such as the present ones, as well as those in the *Principia*) are subject to the theory of types, or they are not.

A. If they were, the theory would include within its own scope propositions of a higher order, and thus be an argument to what is an argument to it.[2]

B. If they were not, there would be an unlimited number of propositions, not subject to the theory, that could be made directly or indirectly about it. Among these propositions there might be some which refer to a totality and involve functions which have arguments presupposing the function.

3. The statement of the theory of types is either a proposition or a propositional function, neither or both.

A. If it were a proposition, it would be either elementary, first order, general, etc., have a definite place in a hierarchy and refer only to those propositions which are of a lower order. If it were held to be a proposition of the last order, then the number of orders would have a last term, and there could not be meaningful propositions made about the theory. The *Principia* should not be able to say, on that basis, just what the purpose, character and application of the theory is.

B. Similarly, if it were a propositional function, it would have a definite place in a hierarchy, being derived from a proposition by generalisa-

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[1] Paradoxes, though contrary to common opinion, may be and frequently are true. Paranoumena, violating principles of logic or reason, if they are not meaningless, are false, and it is only they which are capable of logical analysis and resolution. What the *Principia* attempts to do is to solve apparent paranoumena with a real paranoumenon.

[2] P. 39, *Principia Mathematica.*

tion. It could not refer to all propositions or propositional functions, but only to those of a lower order.

C. If it were neither it could not be true or false, nor refer to anything that was true or false. It could not apply to propositions, for only propositions or propositional functions, in a logic, refer to propositions.

D. If both at once, it would be necessarily self-reflexive.

a. If as function it had itself as value, it would refer to itself. But the theory of types denies that a function can have itself as value.

b. If as function it had something else as value, it would conform to the theory, which insists that functions have something else as values. The theory then applies to itself and is self-reflexive, and thus does not apply to itself. As, by hypothesis, it is a value of some other function, there must be propositions of a higher order and wider range than the theory of types.

It is no wonder that the perpetrators of the theory have not been altogether happy about it! What is sound in it—and there is much that is—is best discovered by forgetting their statements altogether, and by endeavouring to analyse the problems it was designed to answer, without recourse to their machinery. The result will be an acknowledgment of a theory of types having a limited application, and a formulation of a principle which will permit certain kinds of unrestricted general propositions.

To do this we shall deal in detail with two apparent paranoumena dealt with in the *Principia*, where the difficulty is largely *methodological*. We shall then treat of Weyl's "heterological-autological" problem, where the difficulty is due to a confusion in *meanings*. Those problems which cannot be dealt with under either heading will be those which need a theory of types for their resolution.

1. *Epimenides*. The proposition "All Cretans are liars" must be false if it applies to Epimenides as well, for it cannot be true, and only as false has it meaning. If it were true, it would involve its own falsity. When taken as false, no contradiction, or even paradox, is involved, for the truth would then be "*some* Cretans tell the truth". (The truth could not be "all Cretans tell the truth" for Epimenides must be a liar for that to be true and by that token it must be false). Epimenides himself would be one of the lying Cretans, and one of the lies that the Cretans were wont to make would be "all Cretans are liars". Thus if Epimenides meant to include all his own remarks within the scope of the assertion, he would contradict himself or state a falsehood. If it be denied that a contradictory assertion can have meaning, he must be saying something false if he is saying anything significant. Had he meant to refer to all other Cretans there is, of course, no difficulty, for he then invokes a kind of theory of types by which he makes a remark not intended to apply to himself. All difficulty disappears when it is recognised that the formal implication, "all Cretanic statements are lies" can as a particular statement be taken as one of the values of the terms of this implication. Letting $Ep\ !\ p$ represent "Epimenides once asserted p"; ϕ represent "Cretanic" and p represent a statement or

proposition, then for "All Cretanic statements are false (or lies)," we have:

1. $\phi p \,.\supset_{p}.\backsim p.$

And as Epimenides is a Cretan, for any assertion he makes we have:

2. $Ep \,!\, p \,.\supset_{p}.\, \phi p.$

As No. 1 is an argument to the above—it being Epimenides' present remark—we get:

3. $Ep \,!\, \{\phi p \,.\supset_{p}.\backsim p\} \,.\supset.\, \phi\{\phi p \,.\supset_{p}.\backsim p\}$

No. 1, as a Cretanic statement, is an argument to No. 1 as a formal implication or principle about Cretanic statements, so that:

3A. $\phi\{\phi p \,.\supset_{p}\backsim p\} \,.\supset.\backsim\{\phi p \,.\supset_{p}.\backsim p\}$

No. 3 and No. 3A by the syllogism yield:

3B. $Ep \,!\, \{\phi p \,.\supset_{p}.\backsim p\} \,.\supset.\backsim\{\phi p \,.\supset_{p}.\backsim p\}$

so that in this instance Epimenides lied.

It is important to note that No. 1 states a formal implication, and that No. 3, No. 3A and No. 3B employ No. 1 as a particular assertion or specific argument to their functions. No. 3A is an instance of the implication expressed by No. 1, and is this instance because of the particular argument it does have. It states the fact that " 'all Cretanic statements are false' is a Cretanic Statement," implies that " 'all Cretanic statements are false' is false". Substitution of another argument would give a different instance; though of course of the same implication. The implication contained in its argument does not have instances. " 'Some Cretanic statements are false' is a Cretanic statement" or " 'This Cretanic statement is false' is a Cretanic statement" are not instances of " 'All Cretanic statements are false' is a Cretanic statement," but of "P is a Cretanic statement". These three propositions have different subjects; they are different values of the same propositional function. That these subjects have relations to one another is of no moment. "My wife loves me" and "my mother-in-law is old (or loves me)" are two distinct and logically independent propositions, even though there is a relationship between the two subjects.

It is because any considered general proposition is at once an individual fact, and a formal implication or principle, with many possible arguments, that it is capable of being taken as an argument to itself. All propositions about words, logic, truth, meaning, ideas, etc., take arguments which fall in these same categories, and in so far as such a general proposition is stated in words, determined by logic, etc., it should, as such a fact, be an argument to itself as a formal implication. The principle must be false if this cannot be done, for it is sufficient, in order to overthrow a proposition of this kind, to produce one argument for which it does not hold. One may limit the principle by asserting that it holds for "all but . . .", in which case it is a *restricted* general proposition. Nominalism, association of ideas, scepticism, the theory of universal tautology, the denial of logic are defended in propositions which cannot take themselves as arguments, and which as facts are arguments to contradictory principles. Their contradictory principles therefore hold sometimes at least,

so that these doctrines must be false if they are put forward without restriction, and cannot be universally true, if, in Bradley's words, they "appear".

2. "I am lying"—if it be taken in isolation from all fact—is a meaningless statement. There must be some objective truth that is distorted, and unless it is provided the assertion has no significance. This proposition means either, "I am lying about X"; "I always lie," or "I have always lied". The first can be either true or false without giving rise to any problem, except where "all my assertions" is made an argument to X, in which case it is equivalent to either the second or third formulation. "I always lie" involves the same situation as with Epimenides, and the proposition is false. The supposition of its truth would involve a contradiction; the supposition of its falsity means simply that I sometimes lie and sometimes tell the truth. If what is meant is that "I have always lied" that does not involve a contradiction, for what is intended is a restricted proposition, applying to *all but* the present one. It can be true because it does not apply to all propositions; if it were false, then sometimes I lied and sometimes I did not. In short, there is nothing like a self-reflective universal liar, which is an interesting moral conclusion to derive from a logical analysis. Similarly, there cannot be a thorough scepticism held by the sceptic to be valid.

Prof. Whitehead (to whom I am also indebted for the notation) has pointed out to me that wherever a conjunction of propositions results in a *reductio ad absurdum*, there is no way of determining on logical grounds alone which of the antecedents fails, or is false (though one at least must be). Thus in the case of Epimenides we have:

$$4.\ \{\phi p \,.\supset_p.\backsim p\}\ .\ \{Ep\,!\,p\,.\supset_p.\,\phi p\}\ .\ Ep\,!\,\{\phi p\,.\supset_p.\backsim p\}$$
$$\text{(A)}\qquad\qquad\text{(B)}\qquad\qquad\text{(C)}$$
$$.\supset.\backsim\{\phi p\,.\supset_p.\backsim p\}$$
$$\text{(D)}$$

It is because B and C are in that case assumed to hold, that we can say that A must fail. If the truth of all these antecedents were undetermined, we should have merely the general rule: a *reductio ad absurdum* has as a necessary condition the conjunction of one or more false propositions. Transposition—

$$4'.\ \{\phi p\,.\supset_p.\backsim p\}\ .\supset.\backsim\{\phi p\,.\supset_p.\backsim p\}$$
$$\text{(D)}\qquad\qquad\text{(A)}$$
$$.\vee.\backsim\{Ep\,!\,p\,.\supset_p.\,\phi p\}\ .\vee.\backsim Ep\,!\,\{\phi p\,.\supset_p.\backsim p\}$$
$$\text{(B)}\qquad\qquad\text{(C)}$$

makes it apparent that to deny the conclusion of a *reductio ad absurdum* is to imply that at least one of the antecedents is false.

In connection with the *reductio ad absurdum* involved in the assertions, "I always lie" and "I always doubt," No. 4B reduces to the tautologies: "If I assert p, p is my assertion," and "If I doubt, the doubt is mine". In these cases, the only alternatives left are the denial of the fact of the assertion (No. 4C), or the truth of the principle itself (No. 4A).

3. Weyl's heterological-autological contradiction[1] is the result of a material fallacy of amphiboly in connection with the employment of adjectives. The simplest form of such a fallacy is due to a failure to distinguish between an adjective as substantive and an adjective as attribute. Thus if we treat both the subject and attribute in "large is small" and "small is large" as attributes united by a copula expressing identity (instead of reading it as "large is a small word," "small is a large word") we could say "whatever is small is large, and whatever is large is small". No one, I believe, since the Megarics, has been troubled by this particular confusion.

The present problem is the result of a confusion, not between substantive and adjective, but between an adjective which expresses a property, and an adjective which expresses a relation between this property and the substantive. All words can be described in terms of a property—they are long, short, beautiful, melodious, etc., words. They can be classified in accordance with these properties, giving us the class of long words, short words, etc. They can also be classified as either "autological" or "heterological," depending on whether or not the same word is at once substantive and property-adjective; the terms 'autological" and "heterological" expressing relationships between the substantive and adjective.

The autological class is made up of words, each of which expresses a property which it possesses; though all of them have unique properties. If "short" be short, and if "melodious" be melodious, they would both be members of the autological class; though in addition, "short" would be a member of the class of short words, and "melodious" would be a member of the class of melodious words.

The heterological class is made up of words, each of which expresses a property which it does not possess. If "long" be short, and if "fat" be thin, they would both be members of the heterological class; although here also "long" would be a member of the class of short words, and "fat" would be a member of the class of thin words. Though when classified according to the relationship of the adjective to the substantive, "short" would be an autological word and "long" a heterological word, they would both be members of that class which was defined in terms of the properties of words—being in this case, members of the class of short words.

Now if heterologicality were a property that a word could have, and if the word "heterological" had that property, it would be a member of the autological class, for it would then possess a property that it expressed. But it would also be a member of a class of words which had the *property* of heterologicality. This class is determined by taking the properties of words, and if it be called

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[1] Briefly stated it is: all words which express a property they possess are autological; all words which express a property they do not possess are heterological. If 'heterological' is heterological it expresses a property it possesses and is thus autological; if it is autological, it expresses a property it does not possess and is therefore heterological. *Das Kontinuum*, p. 2.

"heterological," must be distinguished from that class which was determined not by properties, but by the relationship between properties and substantives.

If there were a property like autologicality and if "heterological" had that property,[1] it would be a member of the heterological class, for it would express a property which it did not possess. But it would also be a member of the class of words which possessed autologicality and could thus be classified.

Thus if "heterological" had the property of autologicality, it would be in the heterological class owing to the *relation* which held between the property and substantive (or between a property it possessed and the property it expressed); but it would be in the class of autological words, owing to a *property* it possessed. If it had the property of heterologicality, it would be in the autological class on the basis of the *relation*, and in the class of heterological words on the basis of *property* classification. There is no difficulty in considering something as a member of two distinct classes, owing to the employment of different methods of classification. There is no contradiction in saying: " 'heterological' expresses the property heterologicality, possesses the property autologicality, and the relation between these properties is heterological, or that it expresses and possesses the property heterologicality and the relation between them is autological." Similarly, Richard's contradiction, Berry's contradiction, and that involving the least indefinable ordinal, are resolvable by recognising that "nameable" and "indefinable" are used in two sharply distinguishable senses. They do not require a hierarchy, but a discrimination in the methods of description.

When a distinction is made between a class and its membership (the distinction between a number of numbers and a number is a particular case of this), and between a relation of objects and a relation of relations, the requirements for the solution of the other mathematical problems are provided. A class is other than its members, and a relation, like all universals, transcends any given instance or totality of instances. As they have characters of their own, universals can be described in terms of other universals, which in turn transcend them. Arguments are of a different "type" than functions, just so far as they have different logical characteristics, *i.e.* are different kinds of logical facts. The class which is an argument to a function about classes has, as argument, a different logical import than the function, and its arguments have a different import from it. This is true of all functions, restricted and unrestricted alike, for it means simply that they are discriminable from their arguments. They can, despite this difference, have characteristics in common with their arguments, and are to that extent unrestricted. Thus in the case of "the class of those classes which are identical with themselves," the class of classes can be

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[1] 'Heterological,' in fact, has the properties of being long, polysyllabic, etc., and it is questionable whether there are properties like autologicality and heterologicality possessed by words. If there be no such properties, "heterological" is a member of the class of long words, polysyllabic words, etc. In addition it would be one of the terms related by the heterological relation, which fact would not make it have the *property* of heterologicality.

taken simply as a class, without logical embarrassment. Yet a class of classes differs from a class, and must therefore be capable of a different characterisation, and thus also be an argument to a function of a different type. With some classes, it may not be possible to consider them as arguments to their own functions, without uncovering a contradiction. In such cases (*e.g.* the class of those classes which are not members of themselves, and the relations which are connected by their contradictories), it is the difference between the function and the argument that is of moment. That *some* functions cannot take themselves as arguments does not indicate that *all* functions are restricted in scope, but simply that they are *non-restricted*. Some classes and functions are restricted and some are not. To say that all are restricted because some are is an obvious fallacy.

Whenever, as individual, a general proposition is in the class of those objects of which it treats, but cannot be considered as an argument to itself, it is either false or restricted in scope. If the second, its range of arguments must be specified. Accordingly, we can state as a *necessary* condition for the truth of a general proposition, whose scope is unspecified, that when it has a character, which is one of the characters about which it speaks, it *must* be an argument to itself. Thus if Bergson adequately described the comic, his formula should be an object of laughter, and if the theory of types is universal in application, it should be capable of being subject to itself. Conformity to this condition indicates that the unrestricted proposition is *possibly* true; not that it is necessarily true. To demonstrate that such a proposition was necessarily true, it would be essential to show that the supposition of its falsity assumes its truth. That there is danger in applying this rule can be seen from the consideration of some such proposition as: "Everything is made up of language elements". Its denial will be made up of language elements, and would seem to demonstrate that the proposition was necessarily true. Supposition of the falsity of a proposition, however, means verbal denial only in so far as the proposition applies to the realm of language. If it applies to everything, supposition of its falsity involves the positing of the objects of assertions; not the assertions. A necessary unrestricted proposition about everything can be supported only by a demonstration that the supposition of an argument for which it does not hold is self-contradictory. If the proposition has to do with grammar, meaning, logic, judgment, etc., the conditions for a necessarily true and unrestricted proposition would be: 1. the assertion of it is an argument to it; 2. any possible denial is an argument to it. That "any possible denial" rather than "any given denial" is required, is apparent from the consideration of the following propositions: "All sentences are made up of eight words," "No sentence is made up of eight words". Each of these contains eight words. It is because of the fact that we can formulate propositions such as, "It is false that every proposition must be made up of eight words," that the condition is seen not to have been met.

An unrestricted proposition applies to every member of the category, and has some aspect of itself as value. It is in some sense then a determinate in the category which it determines. If the proposition refers to some other category

than the one to which it as fact, or some aspect of it as fact, belongs, it is restricted. Thus "all men are mortal" is neither man nor mortal, and as condition does not determine itself as fact. Any proposition referring to that statement would be of a different type, and would deal with its truth, falsity, constituents, historical place, logical structure, etc. Though the unrestricted propositions have no limitations, the category to which they refer may have. Epimenides' remark, for example, referred only to Cretans. As his assertion was a determinate in the category, and as his statement of the supposed conditions imposed on the members of that category was not a possible argument to the general proposition, the general proposition was seen to be false or restricted. Had he said, "All Cretans tell the truth," he would have stated an unrestricted proposition which was possibly true. It could not be said to be necessarily true unless Cretans and lie, against the evidence of history, were actually contradictories.

Accordingly, we shall say: *All true unrestricted propositions are arguments to themselves; or by transposition, those propositions which are not arguments to themselves are either restricted or false.* As this proposition can take itself as argument it is possibly true. Unless no proposition is possible which does not conform to it, it cannot be said to be necessarily true. I have not been able to demonstrate this and therefore accept it as a definition or "methodological principle of validation". The theory of types, in its most general form, may be stated as: *A proposition or function of order n, which cannot be an argument to itself, is, as fact, an argument of a proposition or function of order n+1.*

In accordance with the scheme of the criticism of the theory of types, we can describe our principle as (1) applying to all propositions, including (2) those which refer to it. (3) It is a formal implication with itself as one of its arguments. The theory of types, on the other hand, (1) does not apply to all propositions, but only to those which are restricted, (2) *may* apply to those propositions which refer to it, and (3) is a formal implication which cannot take itself as argument.

The theory of types cannot be an unrestricted proposition about all restricted propositions. As an unrestricted proposition it must take itself as argument; but its arguments are only those propositions which are *not* arguments to themselves. It cannot therefore be unrestricted without being restricted. Nor can it be a restricted proposition about all restricted propositions for it would then be one of the restricted propositions, and would have to take itself as argument—in which case it would be unrestricted. Hence it cannot be restricted without being unrestricted. Three possible solutions may be advanced. The first is that the theory of types is restricted and does not apply to *all* restricted propositions, but only to *some* of them. It is not an argument to itself but to some other proposition about restricted propositions. This in turn will have to be restricted and refer only to some propositions, and so on, giving us theories of types of various orders. The proposition made about the totality of these orders would be of a still higher order and would in turn presuppose a higher order *ad infinitum.* The theory of types thus depends on theories of types of theories of types without end. This seems probable on the

ground that the theory is based on the recognition that no proposition can be made about all restricted propositions, so that it must by that very fact admit that it cannot apply to all of them. Instead, therefore, of the theory of types applying to all propositions, and determining them in various orders, it does not even apply to all of a given class of them. This interpretation would not affect unrestricted propositions, and would merely show that the determination of restricted propositions is subject to determinations without end.

The second possibility is suggested by the consideration of a proposition such as: "all truths are but partially true". If that were absolutely true, it would contradict itself, and if it were not, could apply only to some truths. Considered as referring to the necessary limitations which any finite statement must have, it would take itself as argument in so far as it was finite, thus indicating that it was absolutely true about finite propositions, and yet not absolutely true as regards all truths. By pointing out the limitations of a finite statement it indicates that there is an absolute truth in terms of which it is relatively true. On this interpretation, any condition which imposes universal limitations is unlimited in terms of what it limits, but limited in turn by some other condition. One might hold, therefore, that the theory would be unrestricted as regards restricted propositions, and restricted as regards all propositions, and would point to a higher principle which limits it.

The third possibility is to allow for "intensive" propositions which are neither restricted nor unrestricted, being incapable of any arguments. The theory of types could be viewed as such an intensive proposition, and what we have called its arguments, would merely "conform" to it. This interpretation means the downfall of a completely extensional logic, and a determination of an extensional logic as subordinate to an intensional one.

There are difficulties in each of these interpretations. The last seems to me to be best. In any of these cases, however, a restricted proposition which refers to some other than the restricted aspect of the theory would be subject to the theory and the principle we have laid down about unrestricted propositions could still hold. Those restricted propositions which refer to the restricted character of the theory would not be an argument to it on the first, would be an argument to it on the second, and would neither be nor not be an argument to it on the third solution.

To briefly summarise: The theory of types must be limited in application. Not all the problems it was designed to answer require it; another principle of greater logical import is desirable; while for the resolution of the problems in which it is itself involved, very drastic remedies are necessary. No matter how the theory fares, the possibility of the methodological principle and the possibility of other solutions for the so-called paradoxes, indicate that it is at least not as significant an instrument as it was originally thought to be.

SUPPLEMENT III

A NON-ARISTOTELIAN SYSTEM AND ITS NECESSITY FOR RIGOUR IN MATHEMATICS AND PHYSICS*

BY ALFRED KORZYBSKI

We are here dealing with a concrete mathematical problem which is not trivial, but at the same time is solvable, and I cannot imagine that any mathematician can find the courage to elude its honest solution by means of a metaphysical dogma. (549) HERMANN WEYL

I protest against the use of infinite magnitude as something completed, which in mathematics is never permissible. Infinity is merely a *façon de parler*, the real meaning being a limit which certain ratios approach indefinitely near, while others are permitted to increase without restrictions. (74) K. F. GAUSS

A very extensive literature shows that the problems of 'infinity' pervade human psycho-logical reactions, starting from the lowest stage of human development up to the present and that without some theory of 'infinity', modern mathematics would be impossible. Up to date, no satisfactory theory of infinity, on which all mathematicians could agree, has been produced. The results are rather bewildering because what appears to some prominent mathematicians as perfectly sound mathematics is evaluated by other equally prominent scientists as a 'mental' disease (Poincaré); or we find opinions that a large portion of mathematics is devoid of proof and has to be accepted on faith; or that some parts of mathematics must be treated as non-sense (Kronecker, Brouwer, Weyl . ,). 'There are eminent scholars on both sides and the chance of reaching an agreement within a finite period is practically excluded', says Brouwer, and certainly such a state of affairs does not allow us to have any satisfactory modern standards of proof and rigour; the last thing we should expect in mathematics.

The majority of those mathematicians who take interest in the soundness of their science seem to believe that the main difficulty centres around the validity of the 'law of excluded third' ('A is B, or not B') of the accepted, sharply two-valued, chrisippian form of *A* 'logic'. They disregard the fact that we are born, bred, educated, speak a language, live under conditions, institutions . , which still remain desperately *A* or even pre-aristotelian. If we attempt to reject one of the two-valued 'laws of thought' or postulates of the *A*-system, but retain *A* or pre-aristotelian *elementalistic* 'psychologies', 'logic', and *s.r*, no agreement in 'a finite period' can be expected, and the present mathematical chaos would continue.

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*Paper presented before the American Mathematical Society at the New Orleans, Louisiana, Meeting of the A.A.A.S. December 28, 1931. I continue to use the abbreviations introduced in this book.

Among the more important schools we may distinguish roughly:[1]

1) The logistic school represented by Peano, Russell, and Whitehead, who accept the chrisippian, two-valued, restricted form of the *el* 'logic' and so may be called the *chrisippian school.*

2) The axiomatic school, represented by Hilbert and his followers, which may be called the *aristotelian school.*

3) The 'intuitional' school represented by Brouwer and Weyl who question the 'law of excluded third', and so may be called the *non-chrisippian school.*

4) The Polish school of: (*a*) 'intuitional' formalism with Łukasiewicz, Tarski, Leśniewski as representatives, which may be called the *non-aristotelian school.* Łukasiewicz generalized the A 'logic' to three-valued 'logic' which covers modality. Łukasiewicz and Tarski finally produced a general many-valued 'logic' of which the two-valued represents only a limiting case. Leśniewski produced *Protothetic*, a still more general 'logical' system, by introducing variable 'funktors', .* (*b*) The *restricted semantic* school represented by Chwistek and his pupils, which is characterized mostly by the semantic approach, and by paying special attention to the *number* of values, establishing the thesis that the older 'freedom from contradictions' depends on one-valued formulations, as discovered by Skarżeński and quoted by Chwistek. This school has already produced new foundations (still *elementalistic*) for 'logic' and mathematics, and leads to generalized arithmetics and analysis.

5) The average prevalent mathematical technician, who does not realize that he belongs to the numerically large class which may be called the 'christian science' school of mathematics, which proceeds by faith and disregards entirely any problems of the epistemological foundations of their supposed 'scientific' activities.

It should be noticed that all existing mathematical schools accept implicitly, at least, A *elementalism* and do not challenge identity, a principle which happens to be invariably false to facts and which therefore should be entirely abolished.

The above classification suggests that, in spite of great achievements in the field of mathematical foundations, no school can expect to be convincing or accepted by other schools as long as we all flounder in the A and *el* ambiguities which prevent any possibility of agreement. It becomes obvious also that when a $\bar{A}$ and *non-el* system is formulated it will necessitate a new paradox-free foundation for mathematics and so a new school of mathematics will arise which may be called:

6) The *general semantic*, *non-aristotelian*, *non-elementalistic* school of mathematics. It is premature to give the names of the leading pioneers in this field at present.

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*At present Łukasiewicz and Tarski call their many-valued 'logic' non-chrisippian, but this name does not seem appropriate because these authors generalized both forms of the aristotelian 'logic' to a many-valued 'logic' of which the two-valued becomes only a limiting case. Thus it seems that their many-valued 'logic' is better described by the term *non-aristotelian,* yet still *elementalistic* 'logic'.

In a $\bar{A}$-system, the 'logical' problems of freedom from contradiction become also semantic problems of *one-valued meanings* made possible only under ∞-valued, $\bar{A}$, *non-el* general semantics, and the recognition of the $\bar{A}$ *multiordinality* of terms , . A $\bar{A}$-system introduces some fundamental innovations, such as completely rejecting identity, elementalism . , and becomes based on *m.o* structure and order, and so ultimately becomes *non-el*. The A, (3+1)-dimensional *el*, (in the main) *intensional* system becomes a four-dimensional, *non-el*, (in the main) *extensional* system. In such a system we cannot use the formulations of *elementalistic* 'logics' and 'psychologies', but must have $\bar{A}$, *non-el* general *semantics*, which when generalized become an entirely general discipline applicable to all life, as well as to *generalized* mathematics. For the above reasons I shall use the word 'logic', in its *el* sense, with quotation marks; and use the term *general semantics* for a *non-el*, $\bar{A}$ discipline corresponding to the *el*, A or $\bar{A}$ 'logics'.

Investigations show that the primitive man (and the 'mentally' ill) use *one-valued semantics* which have left more or less marked traces in all of us, reflected even in science and mathematics. The elimination of these primitive traces clears the foundation for an adult civilization, a theory of sanity, and the elimination of the scientific and mathematical paradoxes.

To assume that because a many-valued 'logic' has been produced, all the problems of mathematical infinity, irrational numbers, continuity, mathematical induction, validity of mathematical proof, mathematical existence . , have been solved, would be a mistake. The aim of the present paper is to analyse some of the fundamental complexities produced by the unconscious operation of the one-valued semantic *identification* concealed in the formulation of the 'law of identity', which have escaped notice until now, and which would make the application of a many-valued 'logic' or ∞-valued semantics and agreement impossible. Here, as in the $\bar{E}$ and $\bar{N}$ systems, only the most general formulations help us to *discriminate* between the particular cases, and so to eliminate the undesirable traces of one-valued semantics by building a $\bar{A}$-system, of which the A and pre-A represent only particular cases.

Let me recall the 'philosophical grammar' of our language which we solemnly call the 'laws of thought', as given by Jevons:[2]

1) The law of identity. Whatever is, is.

2) The law of contradiction. Nothing can both be, and not be.

3) The law of excluded third. Everything must either be, or not be.

These 'laws' have different 'philosophical' interpretations which help very little and for my purpose it is enough to emphasize that: (1) The second 'law' represents a negative statement of the first, and the third represents a corollary of the former two; namely, no third possible between two contradictories. (2) The verb 'to be', or 'is', and 'identity' play a most fundamental role in these formulations. We should not be surprised to find that the investigation of these terms may give us a long sought solution. Such an investigation is very laborious and difficult. 'The complete attempt to deal with the term *is* would go to the form and matter of everything in existence, at least, if not to

the possible form and matter of all that does not exist, but might. As far as it could be done, it would give the grand Cyclopaedia, and its yearly supplement would be the history of the human race for the time', said Augustus de Morgan in his *Formal Logic*, and this opinion I found fully justified.

So I must be brief, and state but roughly, that in the Indo-european languages the verb 'to be' has at least four entirely different uses: (1) as an auxiliary verb, 'Smith is coming'; (2) as the 'is' of predication, 'the apple is red'; (3) as the 'is' of 'existence', 'I am'; (4) as the 'is' of identity, 'the apple is a fruit'. The fact that four semantically entirely different words should have one sound and spelling appears as a genuine tragedy of the race; the more so since the discrimination between their uses is not always easy.

The researches of the present writer have shown that the problems involved are very complicated and cannot be solved except by a *joint study* of mathematics, mathematical foundations, history of mathematics, 'logic', 'psychology', anthropology, psychiatry, linguistics, epistemology, physics and its history, colloidal chemistry, physiology, and neurology; this study resulting in the discovery of a general semantic mechanism underlying human behaviour, many new interrelations and formulations, culminating in a $\bar{A}$-system. This semantic mechanism appears as a general psychophysiological mechanism based on four-dimensional order, present and abused in all of us, the primitive man, the infant, the 'mentally' ill, and the genius not excluded. It gives us an extremely simple means of training our *s.r*, which can be applied even in elementary education.

The scientific problems involved are very extensive and can be dealt with only in a large volume. Here I am able to give only a very sketchy summary without empirical data, omitting niceties and technicalities.

(*a*)

Paris Dresden Warsaw

*_____*_____*

(*b*)

Dresden Paris Warsaw

*_____*_____*

If we consider an actual territory (*a*) say, Paris, Dresden, Warsaw, and build up a *map* (*b*) in which the order of these cities would be represented as Dresden, Paris, Warsaw; to travel by such a map would be misguiding, wasteful of effort , . In case of emergencies, it might be seriously harmful , . We could say that such a map was 'not true'. , or that the map had a *structure not similar* to the territory, structure to be defined in terms of relations and multi-dimensional order. We should notice that:

A) A map may have a structure similar or dissimilar to the structure of the territory. (1)

B) Two similar structures have similar 'logical' characteristics. Thus, if in a correct map, Dresden is given as between Paris and Warsaw, a similar relation is found in the actual territory. (2)

C) A map *is not* the territory. (3)

D) An ideal map would contain the map of the map, the map of the map of the map . , endlessly. This characteristic was first discovered by Royce. We may call it self-reflexiveness. (4)

Languages share with the map the above four characteristics.

A) Languages have structure, thus we may have languages of *elementalistic* structure such as 'space' *and* 'time', 'observer' *and* 'observed', 'body' *and* 'soul', 'senses' *and* 'mind', 'intellect' *and* 'emotions', 'thinking' *and* 'feeling', 'thought' *and* 'intuition'. , which allow verbal division or separation. Or we may have languages of *non-elementalistic structure* such as, 'space-time', the new quantum languages, 'time-binding', 'different order abstractions', 'semantic reactions'. , which do not involve verbal division or separation . ; also mathematical languages of 'order', 'relation', 'structure', 'function', 'variable', 'invariant', 'difference', 'addition', 'division'. , which apply to 'senses' and 'mind', that is, can be 'seen' and 'thought of', . (5)

B) If we use languages of a structure non-similar to the world and our nervous system, our verbal predictions are not verified empirically, we cannot be 'rational' or adjusted , . We would have to copy the animals in their wasteful and painful 'trial and error' performances, as we have done all through human history. In science we would be handicapped by semantic blockages, lack of creativeness, lack of understanding, lack of vision, disturbed by inconsistencies, paradoxes , . (6)

C) Words *are not* the things they represent. (7)

D) Language also has self-reflexive characteristics. We use language to speak about language, which fact introduces serious verbal and semantic difficulties, solved by the theory of *multiordinality*. (8)

The above unusually simple considerations lead to unexpectedly far-reaching consequences.

A) From (7)—it follows that the objective levels which include the events, ordinary objects, objective actions, processes, immediate feelings, 'instincts', 'ideas', *s.r* in general . , represent un-speakable levels, *are not words*. (9)

B) From (9)—that the use of the 'is' of *identity*, as applied to objective, un-speakable levels, appears invariably structurally false to facts and must be entirely abandoned. Whatever we might *say* a happening 'is', *it is not*. (10)

C) From (10)—*structure* appears as the only possible link between the objective, un-speakable, and the verbal levels. (11)

D) From (11)—the only possible 'content of knowledge' becomes exclusively *structural*. (12)

E) From (12)—the only aim of 'knowledge' and science appears as the empirical search for, and verbal formulation of, structure. (13)

F) The only method for acquiring 'knowledge' is found in an *empirical* investigation of the potentially unknown structure of the world, ourselves included, only afterwards adjusting the structure of languages so that they would be similar, and so of maximum usefulness; instead of the delusional

reversed order of ascribing to the world the structure of an inherited primitive language. (14)

G) The investigation of the potentially known structure of languages in which we predict and then verify the predictions empirically, appears as an important method for the discovery of the structure of the world. (15)

H) Investigations disclose that all A, *el* languages and disciplines built on them (older 'psychologies', 'logics'. , and, based on them, economics, sociology, politics, 'ethics'. , reflected in turn in our institutions, systems . ,) *are not* structurally similar to the world and our nervous system, as they verbally divide what empirically cannot be divided. Under such conditions neither a higher grade civilization, nor general sanity, nor paradox-free science and mathematics are possible. In *el* languages, our verbal predictions are not verified empirically, and not being able to foresee we must proceed by animalistic 'trial and error'. (16)

I) Mathematics appears as a very limited but the only language in existence, in the main similar in structure to the world around us *and* the nervous system. (17)

J) From the study of mathematics, mathematical physics, and physics, we learn, and will continue to learn, the fundamentals of *m.o* structure. It is no mystery that all chemistry has become a branch of physics, all physics can be made a branch of geometry, all geometry a part of analysis, and all analysis a part of general semantics. The present work shows that the analysis *of all human problems* of daily life or science becomes dependent on *general semantics* which on the verbal levels becomes generalized mathematics. Thus mathematics, mathematical physics, and physics become the most important disciplines from which we learn most about *structure*,—the only 'content of knowledge'. (18)

K) The older *el* 'psychologies' and 'logics' for their maximum usefulness must be transformed into unified *non-el* psycho-logics and general semantics, possible only after studying all forms of human behaviour, mathematics included. (19)

L) The study of mathematics as a form of human behaviour, appears necessary prior to the possibility of formulating any laws of semantics. (20)

M) The problem of mathematical foundations do not belong to mathematics but to psycho-logics which would not disregard anthropology, and would not be vitiated by our persistence in the use of structurally inappropriate *el* 'psychologies', 'logics', and an innocence of mathematics. (21)

N) The 'intuitional' and the 'intuitional' formalist schools of mathematics must be considered as a legitimate, yet not properly formulated, protest against the older elementalism. (22)

O) The general semantic school will represent the *non-el* and $\bar{A}$ school of mathematics. (23)

P) The present crisis of mathematics ultimately depends on the meanings and use of a few terms such as 'all', 'there is', 'infinite'. , which solution depends on a *non-el* theory of meanings, which ultimately can be solved by transforming

what might be called the (3+1)-dimensional *el*, A-system, which divides 'space' *and* 'time'. , (an attitude which is carried all through the system), into a four-dimensional *non-el*, $\bar{A}$-system (an attitude which is also carried all through the system). (24)

Q) From (8)—it follows that statements about statements represent results of new neurological processes, that their content varies, and that we must *discriminate* and *not identify* these different meanings. In other words, only through consciousness of abstracting which represents the most general *s.r* of discrimination, or the elimination of identification, can we assign single values to words which have an essentially many-valued character. Identification confuses these many meanings into one. (25)

R) We must differentiate between descriptive and inferential words and phrases, and never use inferential terms as descriptive, without realizing that we are doing so. (26)

S) Certain words or phrases used to speak about languages, such as 'all statements', 'proposition about all propositions'. , lead to self-contradictions. We cannot speak about 'all' propositions without some limitations, if we proceed introducing new propositions. Even St. Paul felt the necessity for limiting the values of 'all'.* We are compelled to introduce some equivalents to the biblical 'illegitimate totalities' or the theory of types of Russell. (27)

T) Analysis finds that certain of the most important terms we use; such as, 'yes', 'no', 'true', 'false', 'all', 'fact', 'reality', 'existence', 'definition', 'relation', 'structure', 'order', 'number', 'is', 'has', 'there is', 'variable', 'infinite', 'abstraction', 'property', 'meaning', 'value', 'love', 'hate', 'knowing', 'doubt'. , . , may apply to all verbal levels and in each particular case may have a different content or meanings and so *in general no single content or meaning*. I call such terms *multiordinal terms* (*m.o*). The definition of such terms is always given in other *m.o* terms preserving their fundamental multiordinality. In other words, a *m.o* term represents a many-valued term. If the many values are identified, or disregarded, or confused, we treat a fundamentally many-valued term as one-valued, and we must have every kind of paradox through such an identification. All known paradoxes in mathematics and life can be manufactured by the disregard of this fundamental multiordinality. Vice versa, by formulating the general semantic problem of multiordinality we gain means to discriminate between the many meanings and so assign a single meaning in a given context. A *m.o* term represents a variable in general, and becomes constant or one-valued in a given context, its value being given by that context. Here we find the main importance of the semantic fact established by Skarżeński,** that the 'logical' freedom from contradiction becomes a semantic

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*Professor Cassius J. Keyser drew my attention to a passage in the first letter of St. Paul to the Corinthians, Chapter 15, line 27. 'For he hath put *all* things under his feet. But when he saith *all* things are put under him, it is manifest that he is *excepted,* which did put *all* things under him.' Italics are mine.

**Quoted by Chwistek in his *Neue Grundlagen der Logic und Mathematik.*

problem of one-value. But for application we must have a four-dimensional, *non-el*, $\bar{A}$, extensional system, based on structure . , and the complete elimination of identity. (28)

U) That the disregard of multiordinality, orders of abstractions, may lead to identification and therefore false evaluation resulting in disagreement and maladjustment. (29)

V) From (25–29)—it follows that identification or confusion of higher order abstractions must be eliminated. Because of (7, 9, 10, 25–29)—*all identification must be eliminated.* (30)

W) The elimination of identification on all levels, or a complete and unconscious discrimination between different orders of abstractions, including as a special important case the multiordinality of terms, results in general consciousness of abstracting which in turn, solves the paradoxes of life and mathematics and leads to *generalized* mathematics along the lines suggested by Chwistek. (31)

X) The realization of the inherent multiordinality of some of the most important terms we have, gives us an enormous flexibility of language. It makes the number of our words indefinitely great. When both the writer and the reader recognize this multiordinality, and look for the meaning in the context and discriminate between the orders of abstractions, indicated by the context, confusion becomes impossible. (32)

Y) The test for multiordinality is simple. We take any statement and test it to see whether a given term applies to it. Then we make a statement about this statement and again test if this term applies to the new higher order statement. If it does, the given term must be considered multiordinal, because this procedure may be repeated indefinitely. (33)

Z) *The complete elimination of identification* does not allow us to use the term 'is' of identity, and so we must use operational, functional, actional, behaviouristic . , languages, requiring new attitudes and new *s.r*, impossible without the formulation of a $\bar{A}$-system. (34)

Z_1) The *s.r* of those who produced the general theory of relativity, the unified field theory, the new quantum mechanics, the new revision of the foundation of mathematics . , depend on new $\bar{A}$, *non-el*, and non-identity, operational, actional . , attitudes. (35)

Z_2) As the $\bar{A}$-system is based on the general elimination of the 'is' of identity, or on 'is not', it is impossible to reject these premises without producing impossible data, and a theory of agreement 'in a finite period' then becomes a possibility. (36)

Z_3) The old 'unknowable' becomes abolished and limited to the simple and natural fact that the objective levels *are not* words. (37)

Observation and experience, scientific and otherwise, show that in nature we find a definite order, which establishes a *natural order;* namely, that the sub-microscopic process, called the event or the scientific object, came first; only later abstracting organisms happened and objects which represent the results of abstracting by amoebas or men, came next. In the process of evolution

we find object first, label next. Descriptions first, inferences next. The above *natural order* establishes also a *natural order of evaluation.* Proper evaluation becomes the foundation for survival, *non-el s.r;* the more so since evaluation requires asymmetrical relations of 'more' or 'less'. , impossible to handle properly in an A-system. Thus the most important level is represented by the sub-microscopic processes. What the organism needs is not the three-dimensional shadow of a four-dimensional event, not the abstraction of low order produced by our nervous systems, called the object, but the sub-microscopic dynamic processes without which the desired end-results would not happen. The animal, the primitive, the infant, the ignorant man identify the two; live in a delusional world. Similarly the objective levels are more important than the verbal levels, and descriptions are more important than inferences. If we *identify* any orders while the natural order is established by the asymmetrical relation of 'more', the semantic process of evaluation is *reversed* and appears pathological in different degrees. If $a>b$ and we make them delusionally equal in value (identify), then, in the false-to-fact relation $a=b$ we have either over-evaluated the right-hand side or under-evaluated the left-hand side; in both cases reversing the natural order of evaluation. It is important to notice that by basing our *s.r* on a *natural order* of evaluation, general semantics become a generalized science of order and values, a very secure guide in life, indispensable for sanity, as experiments have shown, and include also generalized mathematics.

Another very serious mechanism of identification is found in language.

A) Thus we have only *one name*, say 'apple' for the: (*a*) un-speakable, un-eatable event or scientific process; (*b*) the un-speakable but eatable abstraction of low order, the object; (*c*) the un-speakable and un-eatable 'mental' picture, or higher order abstraction, on semantic levels; (*d*) and for a definition on verbal levels. (38)

B) The multiordinality of terms was not discovered until 1925 and is still generally unknown. It presents a serious difficulty facilitating, perhaps even necessitating, identification unless prevented by special formulations and training. Multiordinal terms sound and look alike on all levels; experience has shown how easy it is to confuse their orders and identify the many values into one. (39)

C) The differentiation between descriptions and inferences, and particularly between descriptive and inferential words as such, is also novel, and was, until the present $\bar{A}$-system was formulated, largely disregarded, which again led to identifications and false evaluations. (40)

Investigations show, that in all known primitive peoples and in the 'mentally' ill, we find literal identification of different orders of abstractions, which accounts for these semantic states. Even their 'perceptions' are different from those of the so-called 'normal', 'civilized' man, because higher order abstractions are projected and identified with lower order abstractions. They identify or ascribe one value to essentially many-valued different orders of

abstractions and so become impervious to contradictions with 'reality' and impervious also to higher order experience.

The infant, and the rest of us, identify a great deal because of the reasons given above. Investigations show that most of human difficulties, public, private, or *scientific* are due to this *A s.r*, which accounts for the infantile state of our commercial so-called civilization. Identification abolishes the natural order of evaluation, but so does also an unconscious assumption of an 'infinite velocity' of a process. The *A* trilogy involved some fanciful 'infinity' assumptions. Thus in the *A*-system the velocity of nerve currents, which is known to be 126 metres per second in the human nervous system, is at present assumed unconsciously as 'infinite', made evident by the elementalism of 'intellect' or 'emotions'. , as something 'by themselves' and detached. In the *E*-system the length of a line, the space constant, and the natural unit of length were assumed 'infinite'. In the *N*-system the velocity of light, known to be finite, was unconsciously assumed to be 'infinite'. In the $\bar{A}$ trilogy these unjustified or meaningless 'infinities' have been eliminated. 'Infinite velocity' of a process has *no meaning*. It represents only a play upon symbols. Velocity is defined as $v=s/t$. If we assume $t=0$ and write $v=s/0=\infty$, this 'velocity' lacks one of the fundamental factors of its definition; namely, t, and so such an expression ceases to define anything at all and has no meaning, although it may be a symbol for a semantic disturbance. But the results of such delusional *s.r* are far reaching, no matter how mild they might be in degree. In a process propagated with 'infinite' velocity there would be no transition or delay in action, and therefore such a process would *not be ordered*. Vice versa, the disregard of order in our observations must introduce some mythological 'infinities' somewhere. So we see that the semantic process of identification is intimately connected with 'infinity' assumptions, both *abolishing order*. Training in natural order trains *s.r* away from delusional evaluation, abolishes pathological identification of different degrees and fanciful 'infinities'.

Thus we see that the problems of mathematical 'infinity' are extremely complex and involve many fundamental considerations never analysed before in connection with the semantic process of identification. Once these problems are analysed and formulated from a $\bar{A}$, *non-el*, structural point of view the problems of 'excluded third' become secondary in importance, easily managed under the creative freedom of the coveted 'consciousness of abstracting'.

Let me recall for continuity, that the mathematicians recognize at present, two kinds of 'infinities'. One with which we are familiar from our school days, symbolized by ∞, Cantor calls 'potential' infinity and defines as a *variable finite*, the misunderstanding of which introduces paradoxes even in high schools; the other, the 'actual' infinity, which introduces paradoxes in universities. All these paradoxes are due, as the present enquiry shows, to fundamental fallacies in connection with semantic processes of identification which we learn at home and in elementary schools.

The process of identification of different orders of abstractions may be due to pathological conditions, to ignorance, to 'thoughtlessness', to lack of

observation, to unconscious false assumptions, to hastiness, to superficiality, to habits of speech, to the structure of language used , . In fact, under the A-system it is practically impossible to avoid it, as we can witness it in such a comparatively advanced field as mathematics. The label 'identification' is applied to the semantic process of wrong evaluation going on inside of our skins on the un-speakable objective levels, when we are not aware of the differences between different orders of abstractions. When making it conscious, we may speak of the confusion of the orders of abstractions. To make such a process conscious, we must train in the differentiation or discrimination between different orders of abstractions, and distinguish the different orders by actually learning how to order them. Such training results in general *consciousness of abstracting* which is not inborn, nor fully acquired, even in university training, but which requires *special* training. Experiments in this field are extremely encouraging; in a number of cases, pathological individuals have become 'normal' and the 'unchangeable' human nature has been actually changed. *Infantile reactions in adults are abolished*, and this training becomes a general and simple method for prevention of future semantic disturbances of false evaluation which must result in maladjustment.

To stop identification we must discriminate or differentiate to the limit between what appears always as four-dimensional, absolute individual stages of processes and situations on all levels, verbal included. Let us follow briefly such an actual performance. If we realize (7)—we accept (11–15)—and on *structural* grounds reject the elementalism of the A trilogy as expressed in its 'psychology', 'logic', the division of 'space' *and* 'time', . We accept the non-elementalism of the $\bar{A}$ trilogy as expressed in the new terms in the present work and accept also 'space-time', . The difference is very serious in all fields, when carried consistently all through the system. As we actually deal with four-dimensional dynamic processes which must be considered continually different, and with world conditions changing also continually, statements about such structural conditions, in an extensional sense, must be considered as involving variables, generating *propositional functions*, doctrinal or system-functions, funktors , . But propositional functions, which involve variables, are neither 'true' nor 'false', but ambiguous, and to have a proposition we must assign a value to the variable by *at least* permanently, in principle, assigning a date to it. We must also introduce, in principle, and as a semantic attitude, numerical subscripts to our words. Thus 'apple' in the A-system represents a *name* attached to an *intensional definition*, and space-time considerations do not enter. The term is applied to a definition which might be considered as one-valued and permanent. Now obviously such a language and *s.r* are *structurally non-similar* to the world and our nervous system.

If we try to identify a name for a definition, implying permanence, with the objective level which is made up of absolute individuals, and represents ever-changing processes, we must live in a delusional world in which we should expect every kind of paradoxes and psycho-logical shocks.

In a $\bar{A}$-system, for structural reasons, we must retain the general implications of the term 'apple', so we retain the word. We must make our language extensional in principle, and the name 'apple' an *individual name*, by calling it 'apple$_1$', 'apple$_2$', . The combination of letters 'a-p-p-l-e' implying similarities, the subscripts 1, 2 . , implying individual differences, which automatically prevent identification. But this is not enough. Our 'apple$_1$' represents a name applied to an object *and* a process; its *meaning* becomes only one-valued when we assign to it at least a definite date. Thus the objective 'apple$_{1\ (\text{Dec. 1, 1931.})}$' may be a very appetizing affair, and 'apple$_{1\ (\text{Jan. 1, 1932.})}$' an un-edible wet splash. It should be noticed that the fundamental difference between the A and $\bar{A}$ systems turns out to be a difference of semantic attitudes. The scientific facts are not changed. The 'apple' of 'Adam' or our own did not differ in essential characteristics under discussion. In both the A and $\bar{A}$ systems we actually deal, in principle, with *many-valued processes*. The important problem is to adjust the structure of our verbal processes to the structure of the world; hence a $\bar{A}$-system must be made extensional, *non-el*, four-dimensional , . Here once more, as in general semantics, the ascribing of one value (or at least limited to a small range of values in practice), in a given situation (context), eliminates paradoxes and contradictions on the older 'logical' grounds. We should notice that the multiordinal terms must be considered as names for many-valued *s.r*, depending upon the order of abstractions; hence the name *multiordinal*. Names for happenings on the objective levels apply to many-valued processes but should not be considered multiordinal. All the psychologics of the differential calculus, 'space-time', enter here, yet the whole field is covered semantically if we entirely abandon the 'is' of identity. Instead of training in 'allness' and 'isness'—'this *is* this', we shall train in non-allness, and non-isness—'this *is not* this', in connection with a special diagram called the Structural Differential.

Experience and experiments show that the above seems essential for sanity. It is interesting to notice that mathematics has produced a language similar in structure to the human nervous system. Roughly the central part of the brain which we call the thalamus is directly connected with the dynamic world through our 'senses' and with those semantic manifestations which we usually call 'affective', 'emotions'. , all of which manifest themselves as dynamic. The cortex which gives us the static verbal reactions and definitions, is not connected with the outside world directly but receives all impulses through the thalamus. On semantic levels the thalamus can only deal with dynamic material, the cortex with static. Obviously for the optimum working of the human nervous system, which represents a cyclic chain, where the lower centres supply the material for the higher centres and the higher centres should influence the lower, we must have means to translate the static into dynamic and the dynamic into static; a method supplied *exclusively* by mathematics.

With the above considerations we must discriminate between our semantic capacities for *infinite divisibility* of finites, and for the *generation* of infinite postulated processes which by definition *cannot be exhausted*. If we use a three-

dimensional A language and apply such an 'all' to such an infinite process then we simply produce a self-contradiction. If we apply to such a semantic process a four-dimensional 'all with a date', then we have arrested, for the 'time' being, the process, or taken a static cross section of the infinite process at that date; but then we deal with a finite. Once we are constantly conscious of abstracting in different orders, these subtle differences become quite clear and the solutions of the problems of infinity follow a similar path as the older problems of the 'infinitesimal', which also was self-contradictory, unnecessary for mathematics. When treated as a *variable finite* it was satisfactory and sufficient, and has proven to be a most creative notion in mathematics. In the problems of the irrational, continuity . , similar subtle identifications or non-discriminations of *el*, A, three-dimensional terms with $\bar{A}$, *non-el*, four-dimensional terms occur, which once eliminated, clear up not only the paradoxes, but some self-contradictory, often unconscious, postulates of some parts of mathematics.

Lack of space does not allow me to go into further details, except to suggest how some subtle *discriminations* may help to eliminate identification. In my $\bar{A}$-system the differentiation between orders of abstractions on physiological grounds, the introduction of multiordinality of terms, four-dimensional considerations . , as a structural necessity for all languages, makes the theory of types unnecessary.

For a better understanding of the present work we must at least differentiate:

A) Between numerical experience and mathematics.

B) Between languages with content and languages without content.

C) Between creative building of verbal schemes which, for the sake of generality, have no content, called pure mathematics, and the application of these schemes to actual problems, with content, called applied mathematics.

D) Between the contentless mathematics and the investigation of the foundation of mathematics which represents the investigation of the *s.r* of mathematicians and belongs to a future *non-el* psycho-logics with content.

E) Between different forms of complex adjustment which we have in common with the primitive man, and even the higher animals, and reasoning which starts with conscious observations, passing to descriptions and inferences , .

F) Between the dynamic process of relating ('thinking') on the unspeakable semantic levels, and the verbal expression of 'relations'.

G) Between the use of negative terms, disagreement, contradiction; and self-contradiction. In a $\bar{A}$-system contradictions take the form of self-contradictions.

H) Between *el* 'logics' expressed in terms of 'true', 'false', and modality, and the investigations of one-, two-, three-, and ∞-valued *s.r*, which become a *general theory of values*, and which may some day include all human interests.

I) Between the inherent circularity of 'human knowledge', which must start with sets of undefined terms, and so start with some knowledge, and circular definitions or explanations which define or explain nothing.

This list of suggestions is not exhaustive, and in principle appears as inexhaustible. I selected only a few topics of immediate need.

We should also notice that because on objective levels we deal structurally with absolutely individual stages of processes and situations and by necessity we speak in higher order abstractions and generalities and use many multi-ordinal terms (without the use of which no speaking is possible), so *any positive statement* about the objective levels must be only probable in different degrees, which introduces a fundamental and entirely *general* $\bar{A}$ *principle of uncertainty*. Heisenberg's restricted principle in physics appears only as a special case. For structural reasons we must preserve determinism but because of (11–15)—the older two-valued determinism must be reformulated into the ∞-valued determinism of the maximum probability. The einsteinian introduction of non-elementalism in physics has resulted in the automatic elimination of some semantic blockages in the younger physicists. Some of the semantic results and triumphs of science, besides the new quantum mechanics, can be found in the latest (free from identification of the term 'time' with some objectivity) new entropy of Tolman.[3]

To sum up, we find that although the primitive man or the 'mentally' ill may have some reactions of orientation, or capacity for relating, which we have in common with the higher animals, yet these do not involve 'reasoning' in the sense defined before. Thus a boxer, football player . , does a great deal of reflexive relating and wins his match, but this cannot be considered as reasoning in the strict sense as used in a $\bar{A}$-system. If we attempt *to discuss* something with a primitive or 'mentally' ill individual and write down his processes of relating, we would have to conclude that he uses *one-valued semantics of identification* of many values into one, or a semantics of inclusion by which 'everything *is* everything else'. The 'law of contradiction', or any 'excluded' 'third', or '*n*-th', practically never appears in our sense, yet it is complicated by the use of positive and *negative* terms, to which any meanings connected with some identifications of higher orders may be ascribed. Although his prevailing semantic processes appear as a complete and *literal identification*, yet because of the general orienting and relating capacities of organisms and the character of terms used, it would not be easy or profitable to attempt an *el* formulation of his 'laws of thought'. But a semantic formulation, as given above, is very instructive and comparatively simple.

Our existing *el* 'logic', besides the two-valued type of formulation, involves many different 'philosophical' elucidations, which instead of clarifying the status of 'logic', in general, tends only to conceal the important issues involved in *non-elementalism*. The role that identification plays in a given individual appears always as a deciding factor in his adjustment. Unfortunately, at present, the sinister identification is not counteracted but fostered or even induced by the structure of the languages we use, different mythologies . , and our whole educational, economic, social . , systems.

The two-valued A, *el*, three-dimensional 'logic' does not apply to the world of events, to the objective levels . , and, for the reasons already explained,

does not apply to the study of the foundations of mathematics. It applies to a large extent to contentless technical mathematics, including so-called 'formal logic' of that system.

Formalism *when free* from identification becomes a unique comparative tool in search for structure; formalism *with* identification of different orders of abstractions, a symptom of semantic disturbances, often of a morbid character. It should be realized that we may have one-, two-, three-. , many-, and ∞-valued orientations, which with the exception of one-valued, we should utilize when conditions warrant a particular use in a particular case. Thus in mathematics, for the sake of having mathematics as a standard of evaluation, *we select* a sharply two-valued orientation by which in the old language '*A* is *B* or not *B*', *to allow sharp* statements that for instance, 1+1=2. If we would deliberately postulate that 1+1 may sometimes be equal to 2 and sometimes not equal to 2, we would have forms of representation which would apply perhaps more readily to science and life, *but* mathematics as such would be impossible, and we would be deprived of this sharp tool for evaluation.

It is interesting to notice that mathematicians, by the use of two-valued *semantics*, (not 'logic', because an *el* discipline cannot be 'lived through' at all by non-heavily pathological individuals), have produced the most important disciplines. Thus we have, for instance, the theory of 'variance' (the theory of function), the theory of invariance, the differential calculus, the 1, 2, 3, 4, and *n*-dimensional systems, and a host of other verbal structures similar not only to the world, but to the human nervous system. These results give us means not only to enlarge our mastery of the external world, but when generalized into a *non-el*, $\bar{A}$-system, give us the means for the mastery of the inner world, leading toward sanity.

It is amusing to discover, in the twentieth century, that the quarrels between two lovers, two mathematicians, two nations, two economic systems . , usually assumed insoluble in a 'finite period' should exhibit one mechanism—the semantic mechanism of identification—the discovery of which makes universal agreement possible, in mathematics and in life.